**CST4599 – POSTGRADUATE PROJECT DISSERTATION**

**PROJECT TITLE:**

# Designing and Securing Multi-Cloud Architectures: A Practical Implementation of Multi-Cloud Infrastructure with AWS, GCP, and Azure Using Open-Source DevOps and Monitoring Tools

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## Declaration

I hereby confirm that the work presented in this report and in all associated materials is entirely my own. I affirm that this report has been submitted to Turnitin, and I understand and agree that this work is subject to assessment for plagiarism.

**Acknowledgement**

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## ABSTRACT

The rapid adoption of multi-cloud environments has introduced new challenges in managing, monitoring, and securing distributed infrastructure across diverse cloud providers. This research presents a practical framework that integrates AWS, Azure, and GCP using Terraform for infrastructure provisioning, Prometheus and Node Exporter for performance monitoring, Grafana for unified visualization, Wazuh for security monitoring, and HAProxy for load balancing. The framework was deployed on real virtual machines to simulate enterprise workloads, enabling cross-cloud correlation of operational and security metrics. Empirical observations demonstrate that this integrated approach improves threat detection, operational visibility, and performance stability while maintaining scalability and reproducibility. By bridging the gap between theoretical multi-cloud security models and real-world deployment, the study provides a deployable methodology for organizations seeking to implement secure, observable, and resilient multi-cloud operations. Limitations include the scale of deployment, absence of automated scaling, and reliance on synthetic workloads, which provide directions for future research.

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# CHAPTER 1: INTRODUCTION

## 1.1 BACKGROUND

The evolution of cloud computing has fundamentally transformed how modern organizations deploy and manage IT infrastructure. Increasingly, enterprises adopt multi-cloud architectures that combine public cloud services such as Amazon Web Services (AWS), Google Cloud Platform (GCP), and Microsoft Azure. This multi-cloud approach provides scalability, cost-efficiency, disaster recovery, and compliance flexibility, positioning it as a preferred solution for dynamic business requirements (Arora et al., 2024; Gupta & Vadlamudi, 2023).

As cloud environments become more complex, challenges related to orchestration, automation, and security grow in parallel. Managing infrastructure across multiple cloud providers requires consistent provisioning, centralized monitoring, and strong security enforcement (Varadaraj, 2025; Soperla, 2025). The rise of DevOps practices, paired with Infrastructure-as-Code (IaC) tools such as Terraform, facilitates automated workflows that streamline deployment across heterogeneous environments (Koneru, 2025; Bobrenok & Guda, 2025).

Open-source tools like Prometheus and Grafana have become instrumental for real-time monitoring and visualization, enabling initiative-taking anomaly detection and performance optimization within cloud-native environments (Van Vugt & Malik, 2023; Minna & Massacci, 2022). Wazuh further strengthens the security posture by providing intrusion detection, log analysis, and compliance monitoring, while HAProxy enhances resilience and availability through load balancing and secure traffic distribution. Despite these advances, integrating and securing multi-cloud architectures remain complex, with risks such as misconfigurations, inconsistent access controls, and fragmented visibility exposing organizations to security vulnerabilities (Sajay et al., 2019; Sudha, 2021).

This research explores a practical implementation of secure multi-cloud architecture leveraging AWS, GCP, Azure, Terraform, Prometheus, Grafana, Wazuh, and HAProxy. The study aims to provide a vendor-neutral, open-source blueprint for organizations seeking operational efficiency and enhanced security within multi-cloud environments (Edet, 2021; Balakrishanan et al., 2024).

## 1.2 STATEMENT OF THE PROBLEM

As enterprises increasingly adopt multi-cloud cloud architectures to balance cost, flexibility, and performance, they are faced with notable integration and security challenges. Gartner (2024) forecasts that nearly 90% of organizations will implement multi-cloud cloud strategies by 2027, highlighting its growing significance in IT management. Yet, the coexistence of multiple public cloud providers such as AWS, GCP, and Azure alongside on-premises infrastructure creates complex environments that are often difficult to automate, monitor, and secure in a consistent manner (Kumar, n.d.; Varadaraj, 2025). In practice, this complexity can lead to operational inefficiencies and unforeseen vulnerabilities.

Traditional cloud management approaches frequently fall short in addressing the dynamic nature of multi-cloud systems. Manual configuration and deployment processes are prone to errors, which can result in misconfigurations, inconsistent implementations, and security gaps (Pathak & Shankar, 2023; Said, Mostafa, Hassan, & Mostafa, 2022). Furthermore, the lack of integrated monitoring and automation tools across platforms can limit visibility and control, thereby increasing the likelihood of downtime, data breaches, and compliance issues (Gopireddy & Gopireddy, 2022; Schork, Zahid, Pradhan, Kicin, & Schwichtenberg, n.d.).

Open-source DevOps and monitoring tools present a promising avenue to mitigate these challenges; however, many organizations struggle with their effective deployment and integration in multi-cloud cloud contexts. Research indicates that standardized practices and reference blueprints for combining such tools across multiple cloud platforms remain scarce, particularly for teams without extensive cloud-native expertise (Bobrenok & Guda, 2025; Gupta & Vadlamudi, 2023; Van Vugt & Malik, 2023).

This study seeks to address these issues by exploring the practical implementation of a secure, automated, and observable multi-cloud cloud infrastructure using open-source technologies. The central challenge lies in designing a scalable architecture that unifies automation, monitoring, and security controls across AWS, GCP, and Azure. By providing a reference framework, this research aims to reduce operational complexity, improve system visibility, and support secure deployment practices within multi-cloud cloud environments (Koneru, 2025; Rapolu, 2025

## 1.3 RESEARCH OBJECTIVES

The objectives of this research are:

1. To conduct a detailed analysis of multi-cloud adoption patterns across AWS, GCP, and Azure.
2. To analyze the security challenges, risks, and threats faced by organizations deploying multi-cloud infrastructures, considering factors such as interoperability, policy enforcement, compliance, and access control.
3. To understand the benefits of adopting a multi-cloud model, including high availability, cost optimization, vendor flexibility, and operational efficiency.
4. To explore best practices in implementing multi-cloud security using open-source tools for automation, monitoring, load balancing, and access management, with insights drawn from real-world case studies.
5. To examine future trends in multi-cloud security and infrastructure management, considering the rise of AI-powered automation, container orchestration, and evolving regulatory landscapes.
6. To design and propose a practical, secure multi-cloud architecture framework and implementation guidelines, integrating AWS, GCP, and Azure platforms with open-source tools such as Terraform, Prometheus, Grafana, Wazuh, and HAProxy to support secure, scalable, and efficient operations.

## 1.4 RESEARCH QUESTIONS

To guide the research and ensure alignment with the objectives, the following research questions have been formulated:

1. What are the current adoption patterns and practices of multi-cloud and multi-cloud architectures across AWS, GCP, and Azure?

2. What are the primary security challenges, risks, and threats faced by organizations in multi-cloud cloud deployments?

3. What are the measurable benefits of adopting a multi-cloud multi-cloud model in terms of availability, cost, flexibility, and operational efficiency?

4. How can open-source DevOps and monitoring tools be effectively implemented to enhance security and automation in multi-cloud cloud environments?
5. What are the emerging trends in multi-cloud cloud security and infrastructure management, particularly in the context of AI, containers, and regulatory requirements?
6. How can a secure and practical multi-cloud cloud architecture framework be designed and implemented to integrate AWS, GCP, and Azure while leveraging open-source tools for scalable and efficient operations?

## 1.5 RESEARCH MATRIX

| **Research Objective (RO)** | **Research Question (RQ)** |
|---|---|
| Analyze security challenges, risks, and threats | What are the primary security challenges, risks, and threats faced by organizations in multi-cloud cloud deployments? |
| Understand the benefits of multi-cloud adoption | What are the measurable benefits of adopting a multi-cloud model in terms of availability, cost, flexibility, and operational efficiency? |
| Explore best practices in multi-cloud cloud security using open-source tools | How can open-source DevOps and monitoring tools be effectively implemented to enhance security and automation in multi-cloud cloud environments? |
| Examine future trends in multi-cloud cloud security and management | What are the emerging trends in multi-cloud cloud security and infrastructure management, particularly in the context of AI, containers, and regulatory requirements? |
| Design and propose a practical multi-cloud cloud framework | How can a secure and practical multi-cloud cloud architecture framework be designed and implemented to integrate AWS, GCP, and Azure while leveraging open-source tools for scalable and efficient operations? |

## 1.6 ASSUMPTIONS, SCOPE, AND DELIMITATIONS

This research is conducted with the assumption that organizations are increasingly adopting multi-cloud strategies for scalability, reliability, and cost-effectiveness. It is also assumed that the selected cloud platforms (AWS, GCP, and Azure) provide sufficient features, documentation, and integration capabilities necessary for secure and efficient implementation.

The scope of this study is limited to the design, security, and monitoring of multi-cloud infrastructures using a combination of public cloud services and open-source tools. Specifically, the research focuses on integrating key services from AWS, GCP, and Azure while leveraging automation with Terraform, monitoring with Prometheus and Grafana, security and log analysis with Wazuh, and load balancing with HAProxy to enhance visibility, resilience, and protection across distributed environments.

This study does not cover private cloud platforms in depth, nor does it analyze vendor-specific proprietary monitoring or security tools unless used for comparison. The performance testing and cost-benefit analysis are constrained to small-to-medium scale deployments, with large enterprise-scale implementations discussed conceptually but not tested in practice. The research also assumes basic familiarity with cloud computing principles and Linux-based systems.

## 1.7 THESIS ORGANIZATION

This thesis is organized into six chapters:

- **Chapter 1: Introduction** – Provides an overview of the research topic, background, problem statement, objectives, assumptions, scope, and the structure of the thesis.
- **Chapter 2: Literature Review** – Reviews existing studies, theories, and frameworks related to multi-cloud cloud architecture, cloud security, and the use of DevOps and monitoring tools in multi-cloud environments.
- **Chapter 3: Research Methodology** – Explains the approach taken to conduct the study, including architectural design, tool selection, implementation procedures, and evaluation methods.

- **Chapter 4: System Implementation** – Details the step-by-step setup of the multi-cloud cloud environment across AWS, GCP, and Azure, including automation using Terraform, and the deployment of Prometheus and Grafana for monitoring.
- **Chapter 5: Results and Discussion** – Presents the outcomes of the implementation, including system performance, security posture evaluation, challenges encountered, and analysis of monitoring data.
- **Chapter 6: Conclusion and Recommendations** – Summarizes key findings, discusses the significance of the study, and provides recommendations for future improvements and research

# CHAPTER 2: LITERATURE REVIEW

## 2.1 INTRODUCTION

Cloud computing has completely changed how organizations design, deploy, and manage their IT infrastructure. The rapid growth of digital services, data-driven applications, and global business needs has created a strong demand for computing environments that are scalable, flexible, and cost-effective. Traditional on-premises setups often struggle with these demands because they have limited scalability, high upfront costs, and complex maintenance. In contrast, cloud platforms offer on-demand resources, elastic scalability, and easy integration, making them an attractive alternative (Gupta & Vadlamudi, 2023; Kumar & Goyal, 2020).

In recent years, many organizations have turned to multi-cloud and multi-cloud architectures to strike a balance between performance, security, cost savings, and vendor independence. These architectures allow workloads to be spread strategically across multiple cloud providers like AWS, Azure, and Google Cloud, while still connecting to on-premises systems. This setup improves availability, disaster recovery, and compliance with specific industry regulations (Varadaraj, 2025; Soperla, 2025).

However, multi-cloud and multi-cloud environments also bring their own set of challenges. Issues like interoperability, data consistency, governance, and security risks make managing these setups more complicated. To overcome these hurdles, organizations need strong architectural designs, effective DevOps practices, and ongoing monitoring. Open-source DevOps tools help automate infrastructure provisioning, configuration, and deployment, while open-source monitoring solutions provide real-time visibility and enable quick responses to problems (Bobrenok & Guda, 2025; Van Vugt & Malik, 2023).

Despite all the research on multi-cloud and multi-cloud architecture, there are still important gaps to fill. For instance, there isn't much work on how to integrate open-source DevOps and monitoring tools across major clouds like AWS, GCP, and Azure in a vendor-neutral way. Current security frameworks often miss the complexities involved in managing policies and automation consistently across different cloud environments. Plus, there are very few practical, real-world examples showing how these open-source tools can be orchestrated effectively in multi-cloud multi-cloud setups. This study aims to fill these gaps by designing comprehensive, open source based architecture that improves automation, monitoring, and security in multi-cloud cloud environments.

## 2.2 AIM

1. Examine the Growth and Adoption of Multi-Cloud Architectures:
   • Explore the evolution of multi-cloud and multi-cloud strategies in modern enterprise environments.
   • Highlight the business and technical motivations for adopting AWS, GCP, and Azure in combination.
2. Evaluate Open-Source DevOps and Monitoring Tools for Cloud Security and Management:
   • Investigate the role of tools like Terraform, Prometheus, and Grafana in automating deployment and ensuring visibility.
   • Analyze the impact of Infrastructure-as-Code (IaC) and continuous monitoring in securing cloud infrastructures.
3. Identify the Security Challenges in Multi-Cloud Environments:
   • Understand cross-platform risks, including misconfigurations, data leakage, and access control issues.
   • Assess current approaches to secure communication, authentication, and network segmentation across clouds.

## 2.3 APPROACH

To conduct a comprehensive and systematic review on the benefits of cloud computing, the security challenges associated with adopting multi-cloud cloud architectures, and the role of DevOps and open-source tools, a structured methodology was employed. The approach involved targeted literature searches using reputable academic and industry platforms to ensure the inclusion of high-quality and relevant studies. Key sources consulted include Consensus, Elicit, ResearchGate, Statista, Mendeley, IEEE

Xplore, the Middlesex University Library (MDX Library), and Google Scholar. These databases were selected for their relevance, scholarly credibility, and access to peer-reviewed content. Search terms were carefully curated and applied consistently across platforms. Clear inclusion and exclusion criteria were defined to ensure objectivity and relevance, forming the basis for a robust foundation of analysis in this literature review.

### 2.3.1 FRAMEWORK

The PRISMA framework (Liberati et al., 2009) was used to guide this systematic review. It ensures a transparent and rigorous selection process by tracking records identified, screened, excluded, and included. A total of 24 studies were selected, consisting of 83% journal articles, 13% conference proceedings, and 4% white papers. Details of the reviewed sources are:

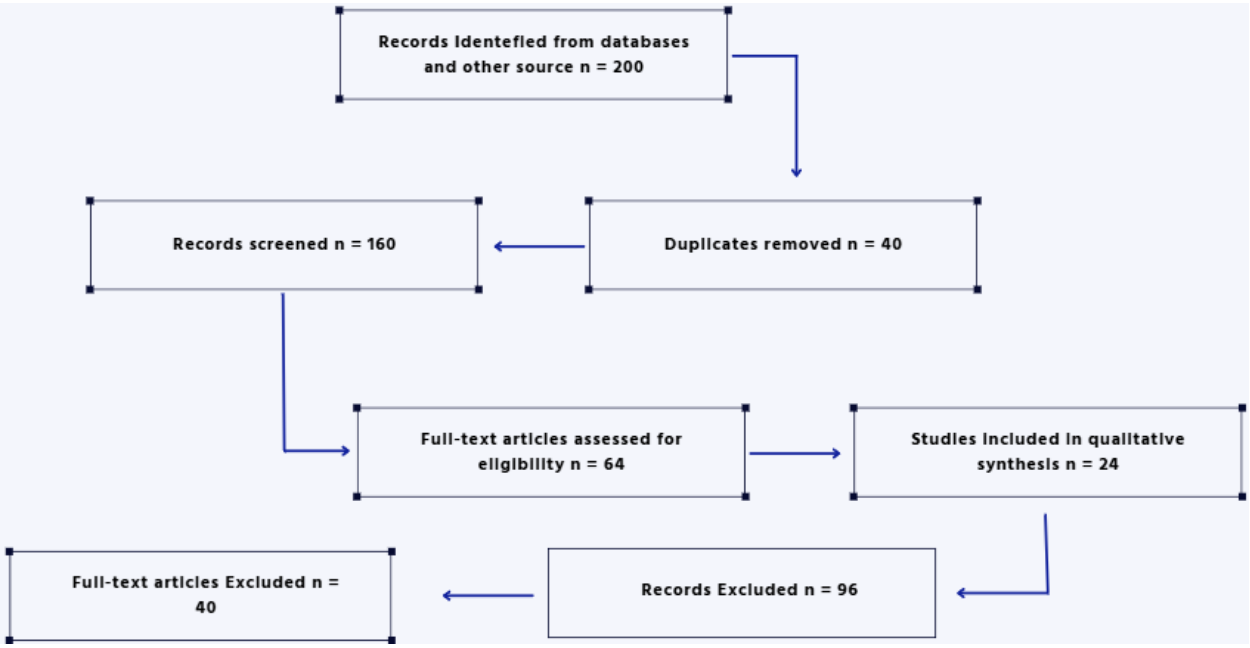



*Figure 1: PRISMA Flow Diagram*

### 2.3.2 DATABASES AND SOURCES USED

In line with the structured methodology outlined in Section 2.3.2 Approach, the literature search was conducted across multiple reputable academic and industry platforms to capture a wide range of high-quality and pertinent studies. The key databases and sources consulted include:

- **Consensus** – Utilized for synthesizing evidence-based research findings and expert consensus on cloud computing and security.
- **Elicit** – Employed for efficient literature discovery using AI-assisted search across scientific publications.
- **ResearchGate** – Accessed to identify peer-reviewed articles and connect with authors in the fields of multi-cloud cloud architectures and DevOps.
- **Statista** – Used for gathering industry data and statistical insights relevant to cloud adoption trends and challenges.
- **Mendeley** – Leveraged both as a reference management tool and a source for accessing scholarly publications.
- **IEEE Xplore** – Consulted for authoritative technical and engineering research related to cloud infrastructure and security.
- **Middlesex University Library (MDX Library)** – Used to access a wide array of academic journals and resources available through institutional subscriptions.
- **Google Scholar** – Employed to supplement the search with grey literature, conference proceedings, and additional peer-reviewed articles.

### 2.3.3 SEARCH TERMS AND KEYWORDS

To capture literature relevant to designing and securing multi-cloud cloud architectures, the following search terms were used in combination with Boolean operators to ensure comprehensive coverage of key themes:

- "Muti-Cloud Architecture"
- "Cloud Security" AND "Challenges"
- "DevOps" AND "Cloud Infrastructure"
- "Open-Source Tools" AND "Cloud Monitoring"
- "Multi-Cloud" AND "Security"

- "Cloud Adoption" AND "Security Frameworks"
- "Prometheus" OR "Grafana" AND "Monitoring"
- "Terraform" AND "Cloud Automation"
- "Wazuh" AND "Security"
- "HAProxy" AND "Load Balancing"

These keywords reflect the core aspects of the research, focusing on the integration of multi-cloud environments, security challenges, the application of DevOps practices, and the use of open-source monitoring and automation tools.

### 2.3.4 INCLUSION CRITERIA

To maintain the relevance and quality of the literature review, the following inclusion criteria were applied:

1. **Publication Date:** Only studies published within the last five years were considered, ensuring coverage of recent developments and current trends in multi-cloud cloud architecture and security.
2. **Language:** Only publications written in English were included to ensure clear comprehension and analysis.
3. **Relevance:** Studies that specifically address multi-cloud and multi-cloud architecture design, cloud security challenges, DevOps implementation, and the use of open-source monitoring and automation tools were selected.
4. **Type of Publication:** The review included peer-reviewed journal articles, conference proceedings, industry reports, and white papers to capture a broad yet credible range of insights.

### 2.3.5 EXCLUSION CRITERIA

To ensure the focus and rigor of the literature review, the following exclusion criteria were applied:

1. **Publication Date:** Studies published more than five years prior to the search were excluded to maintain currency.
2. **Language:** Publications not written in English were excluded.
3. **Relevance:** Studies that do not specifically address multi-cloud cloud architectures, cloud security, DevOps, or related cloud adoption challenges were excluded.
4. **Type of Publication:** Opinion pieces, essays, blog posts, and other non-peer-reviewed or non-technical sources were excluded to preserve scholarly quality.

### 2.3.6 DATA EXTRACTION

For a systematic and comprehensive analysis, a standardized data extraction process was followed. This involved identifying and recording key information from each selected study. The key information extracted from the sources is depicted on the table below.

| **Information Area** | **Description** |
|---|---|
| Bibliographic Information | • Title of the study<br>• Authors and their affiliations<br>• Year of publication<br>• Source (journal, conference, report, etc.) |
| Study Summary | • Research objectives and questions.<br>• Study design and methodology.<br>• Sample size and characteristics (if applicable)<br>• Data collection methods (surveys, interviews, experiments, etc.) |
| Findings and Results | • Key findings related to the design and security of multi-cloud cloud architecture.<br>• Challenges identified in adopting and integrating multi-cloud infrastructure.<br>• Insights on the application of DevOps practices and open-source tools in multi-cloud cloud environments |

| | |
|---|---|
| | • Specific challenges in automating and monitoring multi-cloud cloud security and performance |
| Relevance and Context | • Relevance to multi-cloud and multi-cloud architecture design, cloud security, and DevOps implementation<br>• Context of the study, including industry sectors such as IT, finance, and telecommunications, and geographical considerations where applicable |
| Recommendations and Implications | • Key recommendations for securing and managing multi-cloud cloud environments<br>• Practical benefits for businesses adopting multi-cloud and DevOps tools<br>• Areas suggested for future research in multi-cloud cloud security and automation |

### 2.3.7 DATA SYNTHESIS

The extracted data were synthesized using thematic analysis to identify key themes and insights. The process included the following steps:

1. **Categorization using Collections feature:**
    - Sources were grouped based on similarities and relationships using Mendeley.
    - Categories were reviewed and adjusted to accurately reflect the data.
    - Tags were applied to mark sources excluded from the study.
2. **Identification of Themes:**
    - Major themes were identified from the categorized data.
    - Themes reflected core aspects of multi-cloud cloud architecture design, security challenges, and DevOps implementation.
3. **Refinement of Themes:**
    - Themes were reviewed and refined for clarity and coherence.
    - Sub-themes were created as needed to provide detailed insights.

- Relevant case studies, statistics, and examples were incorporated to support the themes.

## 2.4 SUPPORT RESOURCES

For organizing sources, creating collections, and categorizing journals and research papers the table below shows the distribution of papers by category and year:

| Research Category | Year | Type of Research | Number of Sources | Keywords |
|---|---|---|---|---|
| Open-Source DevOps for Cloud Computing | 2020 | Journal Article | 3 | Open-Source DevOps Tools, DevOps Automation,i Integration/Continuous Deployment (CI/CD) Open-Source DevSecOps Open-Source Cloud-native DevOps |
| | 2021 | | | |
| | 2022 | Journal Article | 3 | |
| | 2024 | | | |
| Open-Source monitoring tools for Cloud Computing | 2023 | Journal Article | 4 | Open-Source Cloud Monitoring, Cloud Infrastructure Monitoring Tools, Cloud Performance Monitoring Open-Source Cloud, Multi-cloud Cloud Monitoring Solutions |
| | 2024 | Journal Article | 3 | |
| | 2025 | Journal Article | 4 | |
| Multi-cloud Cloud Architectures | 2019 | Journal Article, Blog, White Paper | 3 | Multi-Cloud and Multi-cloud Cloud Integration, Multi-cloud Cloud Security Challenges, Data Consistency and Management, |
| | 2020 | Journal Article | 3 | |

| | | | | |
|---|---|---|---|---|
| | | | | DevOps and Continuous Compliance in Cloud |
| Security Challenges in Multi-cloud Cloud Implementation using Opensource DevOps and Monitoring Tools | 2019 | | 1 | Multi-cloud Cloud Security Challenges, Multi-cloud Encryption Algorithms, Access Control Mechanisms, Zero Trust Architecture, Data Confidentiality and Integrity |
| | 2020 | | 3 | |
| | 2021 | | 4 | |
| | 2022 | | 2 | |
| | 2024 | | 5 | |
| | 2025 | | 5 | |
| **Total number of studies reviewed** | | | 33 | |

## 2.5 REVIEW CATEGORIES

### 2.5.1 Existing Cloud Security Architectures and Frameworks

Cloud security architecture has become increasingly vital as enterprises adopt complex cloud solutions. Arora, Lanitha, Muneeswaran, Saha, and Azath (2024) emphasize that advanced security measures, including encryption, access controls, and monitoring, are crucial to protect sensitive data and maintain compliance with regulatory standards. Edet (2021) highlights that security and data privacy are major barriers in multi-cloud adoption, suggesting that structured reference architectures can standardize practices and mitigate risks during the design phase. Balakrishanan, Ahamed, Al-Saidi, and Al-Saidi (2024) demonstrate how scalable architecture incorporating encryption and microservices can enhance security while maintaining operational flexibility. Wang and Ma (2024) extend this perspective by proposing collaborative cloud-network-end frameworks, reflecting a shift from traditional network defenses to cloud-centric, distributed security approaches. Collectively, these studies underscore that while several frameworks exist, there is still a need for comprehensive models that integrate security across multi-cloud and multi-cloud environments.

*Table 2.5.1: Summary of Key Studies on Cloud Security Architectures*

| Author(s) | Year | Key Findings | Gap Identified |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Prakhar Arora, B. Lanitha, Muneeswaran V, Soma Saha, H. Azath | 2024 | Advanced security measures, compliance, addressing data breaches and regulatory compliance | Limited unified frameworks for multi-cloud clouds |
| Henry Edet | 2021 | Reference architecture for multi-cloud security | Needs practical validation and implementation guidance |
| Sreejith Balakrishanan, Z. A. Feroze Ahamed, Sabah Ali Al`abd Al Busaidi, Fatema Khalifa Said Al-Saidi | 2024 | Scalable cloud architectures with encryption, access controls, and microservices | Focused on single-cloud deployment; broader multi-cloud cloud adoption not addressed |
| Xiangyu Wang, Jianfeng Ma | 2024 | Collaborative cloud-network-end security framework; enhances data sharing protection | Broader multi-cloud validation required |

### 2.5.2 MULTI-CLOUD CLOUD SECURITY APPROACHES

Multi-cloud cloud environments face unique security challenges due to their distributed and multi-provider nature. Sajay, Babu, and Vijayalakshmi (2019) propose a multi-cloud encryption algorithm combining homomorphic and Blowfish methods to secure data across clouds. Rashid and Yaseen (2025) examine AI-driven frameworks that dynamically allocate resources and optimize threat detection. Pachala, Rupa, and Sumalatha (2021) highlight multi-cloud multi-cloud approaches using Byzantine protocols, DepSky architecture, and Shamir secret sharing, achieving stronger encryption and authentication than traditional methods like SAML and Kerberos. Sudha and Scholar (2021) present an agent-based framework for comprehensive risk management in third-party cloud environments. Collectively, these studies emphasize that effective multi-cloud cloud security requires multi-layered strategies integrating encryption, intelligent monitoring, and governance.

*Table 2.5.2: Summary of Multi-cloud Cloud Security Approaches*

| Author(s) | Year | Key Findings | Gap Identified |
|---|---|---|---|

| | | | |
|---|---|---|---|
| K. R. Sajay, S. Babu, Y. Vijayalakshmi | 2019 | Multi-cloud encryption algorithm | Focused primarily on storage; integration into full multi-cloud cloud not detailed |
| Mohanad Mohammed Rashid, Omar Mahmood Yaseen | 2025 | AI-driven cybersecurity framework for multi-cloud security management | Limited real-world implementation |
| Sunitha Pachala, C. Rupa, L. Sumalatha | 2021 | Multi-cloud multi-cloud approach using Byzantine protocols, DepSky architecture, Shamir secret sharing | Needs wider experimental validation |
| S. Sudha, O. P. Scholar | 2021 | Agent-based information security framework for multi-cloud cloud risk management | Implementation complexity; requires organizational adaptation |
| Alde Alanda, H. Mooduto, Ronal Hadi | 2023 | Wazuh-based real-time monitoring of AWS EC2 cloud servers, including log collection, malware detection, and active response automation | Requires broader validation across diverse cloud platforms and longer operational periods |
| Candra Kurniawan, Agung Triayudi | 2024 | Real-time detection of gambling web defacement attacks on AWS and GCP using Wazuh and Velociraptor, with Telegram-based alerting and artifact collectio | Requires deeper investigation into inconsistent network isolation outcomes across cloud platforms, particularly GCP |

### 2.5.3 CHALLENGES IN MULTI-CLOUD CLOUD ARCHETECTURES

Despite the benefits of multi-cloud cloud systems, enterprises encounter operational and security challenges. Soperla (2025) highlights the complexity of real-time data integration, including

synchronization, event-driven processing, and secure communication. Varadaraj (2025) emphasizes consistency issues across AWS, GCP, Azure, and on-premises systems, noting integration, security, and governance obstacles. Gupta (2025) identifies interoperability and compliance hurdles despite scalability and cost advantages. Hajari, Paripati, Shah, Prasad, and Narukulla (2020) focus on multi-cloud cloud-edge architectures for AI-driven applications, stressing secure, high-capacity processing in IoT-intensive environments. Malti, Hakem, and Benmammar (n.d.) propose multi-cloud multi-objective optimization algorithms for task scheduling, demonstrating the need to balance efficiency and security. Together, these studies show that effective multi-cloud cloud adoption requires careful architectural planning to address integration, security, and operational risks.

Wasan Saad Ahmed, Ziyad Tariq Mustafa Al-Ta'i, Tamirat T. Abegaz, Ghassan Sabeeh Mahmood (2024) present evidence collection and consolidation across multi-account AWS environments using a combined security lake and modern data lake architecture but call for broader validation across heterogeneous cloud providers and forensic use cases beyond academic environments.

Syed Moiz, Abdul Majid, Abdul Basit, Mansoor Ebrahim, Abdul Ahad Abro, and Mehwish Naeem (2024) demonstrate that cloud-based Wazuh deployment enables effective detection and mitigation of cyber threats, including port scans, DoS attacks, and Metasploit patterns, through streamlined, user-friendly security customization, yet note the need for further validation in large-scale enterprise environments and across diverse organizational security postures.

*Table 2.5.3: Summary of Challenges in Multi-cloud Cloud Architectures*

| Author(s) | Year | Key Findings | Gap Identified |
|---|---|---|---|
| Vijaya Bhaskara Reddy Soperla | 2025 | Real-time data integration; event-driven architectures and stream processing | Implementation guidance limited for enterprises |
| Prabhu Govindasamy Varadaraj | 2025 | Maintaining consistency across AWS, GCP, Azure, and on-premises | Multi-cloud scalability and data governance challenges |
| Sandeep Gupta | 2025 | Interoperability and compliance in multi-cloud cloud integration | Emerging AI-based management trends not fully addressed |

| | | | |
|---|---|---|---|
| Venudhar Rao Hajari, L. Paripati, Jigar Shah, Nitin Prasad, Narendra Narukulla | 2021 | Multi-cloud cloud-edge architectures for AI-driven applications; network capacity and security | Security optimization for IoT-heavy deployment organizational adaptation |
| Arslan Nedhir Malti, M. Hakem, B. Benmammar | 2020 | Multi-cloud multi-objective optimization algorithm for task scheduling in cloud system | Balancing performance and security in distributed environments |
| Wasan Saad Ahmed, Ziyad Tariq Mustafa Al-Ta'i, Tamirat T. Abegaz, Ghassan Sabeeh Mahmood | 2024 | evidence collection and consolidation across multi-account AWS environments using a combined security lake and modern data lake architecture | broader validation across heterogeneous cloud providers and forensic use cases beyond academic environments |
| Syed Moiz, Abdul Majid, Abdul Basit, Mansoor Ebrahim, Abdul Ahad Abro, Mehwish Naeem | 2024 | Cloud-based Wazuh deployment enables effective detection and mitigation of cyber threats, including port scans, DoS attacks, and Metasploit patterns, through streamlined, user-friendly security rule customization | Need further validation in large-scale enterprise environments and across diverse organizational security postures |

### 2.5.4 OPEN-SOURCE SECURITY AND MONITORING TOOLS IN CLOUD INVIRONMENTS

The adoption of open-source security tools has become essential for safeguarding multi-cloud cloud systems. Bobrenok and Guda (2025) evaluate solutions such as Checkov, Falco, and Keycloak, demonstrating their effectiveness in protecting resources while maintaining operational flexibility. Van Vugt and Malik (2023) analyze cloud-native security tools for Kubernetes, showing how they mitigate container-level threats and command-and-control attacks. Kumar and Goyal (2020) propose the ADOC model, a continuous DevSecOps framework integrating security across development, operations, and deployment to ensure secure application delivery. Gupta and Vadlamudi (2023) highlight risks associated

with widespread open-source adoption, citing vulnerabilities like SolarWinds and Log4j, and suggest strategies for secure implementation. Minna and Massacci (2022) develop an open-source testbed for cloud security experiments, enabling practical evaluation of security mechanisms. Ghazizadeh, Tamm, and Creutzburg (2024) present automated tools for cloud security testing, providing frameworks to assess and mitigate multi-cloud cloud risks. Collectively, these studies indicate that while open-source tools are vital for securing cloud systems, effective integration demands expertise, governance, and continuous monitoring.

*Table 2.5.4: Summary of Open-Source Security Tools*

| Author(s) | Year | Tool/Framework | Gap Identified |
|---|---|---|---|
| VV Bobrenok, AI Guda | 2025 | Checkov, Falco, Keycloak | Integration across multi-cloud environments not fully explored |
| Thomas Martijn van Vugt, Tania Malik | 2023 | Cloud-native Kubernetes security tools | Focused on container-level threats; broader multi-cloud cloud deployment not addressed |
| Rakesh Kumar, Rinkaj Goyal | 2020 | ADOC DevSecOps model | Requires organizational readiness and agile adoption |
| Sagar Gupta, S. Vadlamudi | 2023 | Open-source software security policies | Limited experimental validation; risks like SolarWinds and Log4j highlighted |
| Francesco Minna, F. Massacci | 2022 | Open-source cloud testbed for security experimentation | Not yet widely adopted; needs broader testing |
| Hamid Ghazizadeh, Gerrit Tamm, Reiner Creutzburg | 2024 | Automated cloud security testing tools | Deployment guidance limited for multi-cloud cloud integration |
| Aniruddha Ghosh, Sudhanshu Srivastava, P. Supraja | 2024 | Terraform (with Custom Wrapper) | Wrapper) |

| | | | |
|---|---|---|---|
| | | | Limited standardized integration for multi-cloud security and monitoring |
| Perumalsamy Ravindran | 2025 | Terraform, IaC | Limited exploration of unified multi-cloud security and monitoring integration despite efficiency gains |
| Sandeep Pochu, Sai Rama Krishna Nersu, Srikanth Reddy Kathram | 2024 | Terraform, Kubernetes | Limited focus on integrated security and monitoring in multi-cloud DevOps frameworks |

## 2.6 RESEARCH GAP

Although extensive studies exist on cloud security and multi cloud deployment, several gaps remain. Most focus on single-cloud or partial multi-cloud setups (Arora et al., 2024; Sajay et al., 2019), with limited guidance on integrated multi-cloud frameworks that unify automation, monitoring, and security across AWS, GCP, and Azure. Many approaches are theoretical or simulation-based (Pachala et al., 2021; Minna & Massacci, 2022), lacking real-world validation. While open-source tools such as Terraform, Prometheus, Grafana, and Keycloak (Bobrenok & Guda, 2025; Kumar & Goyal, 2020) are promising, standardized integration practices are scarce. Emerging challenges like AI-driven automation, cloud-edge integration, and compliance are often addressed separately (Rashid & Yaseen, 2025; Hajari et al., 2020). This study aims to fill these gaps by proposing a practical, secure multi-cloud cloud architecture using open-source tools across multiple cloud platforms.

## 2.7 CHAPTER SUMMARY

examines multi-cloud and multi-cloud architectures, operational challenges, DevOps strategies, and open-source tools for automation and monitoring. The chapter begins by reviewing existing cloud architectures, highlighting approaches for integrating multiple cloud platforms, ensuring interoperability, and maintaining system efficiency. Multi-cloud cloud deployment challenges, including data consistency,

workflow automation, real-time monitoring, and orchestration, are explored to identify practical barriers to effective multi-cloud implementation.

Open-source tools are evaluated in the context of their ability to support operational efficiency. Terraform is discussed for automation and consistent configuration, Prometheus and Grafana for monitoring and visualization, and Keycloak for identity and access management. Security considerations, while addressed, are treated as one component among several necessary for successful multi-cloud operations.

The chapter concludes by identifying gaps in existing research, noting the lack of practical, integrated frameworks that combine automation, monitoring, and operational management across AWS, GCP, and Azure. These insights directly inform the methodology and framework presented in Chapter 3, emphasizing practical deployment, operational efficiency, and tool integration rather than security alone.

## 2.8 CONCLUSION

The rapid evolution of cloud computing and the widespread adoption of multi-cloud environments have introduced significant security complexities that traditional architectures are ill-equipped to manage. This literature review highlights critical findings directly relevant to designing and securing multi-cloud infrastructures using AWS, GCP, and Azure, combined with open-source DevOps and monitoring tools.

Organizations are actively pursuing multi-cloud strategies to leverage scalability and flexibility but face ongoing challenges in securing data, ensuring consistent identity and access management, and managing interoperability across providers. The growing demand for automation and continuous monitoring has driven adoption of open-source tools such as Terraform, Prometheus, Grafana, Wazuh, and HAProxy to improve deployment efficiency, strengthen threat detection, optimize traffic distribution, and enhance observability.

However, securing these environments requires robust, unified frameworks that address risks like misconfigurations, data leakage, and compliance gaps across disparate platforms. Emerging trends emphasize integrating security into DevOps pipelines (DevSecOps) and leveraging advanced monitoring and security platforms such as Wazuh to detect, analyze, and respond to threats in real time, while employing HAProxy for resilient load balancing and secure access to distributed services.

Building on these insights, this research will propose a practical framework for multi-cloud security that integrates design principles with open-source automation, monitoring, and load balancing solutions. It

will also identify implementation challenges and provide recommendations to ensure resilient, scalable, and secure multi-cloud deployments.

## CHAPTER 3: METHODOOGY

3.1 Introduction

This chapter presents the methodology employed to design, implement, and secure a multi-cloud cloud architecture integrating Amazon Web Services (AWS), Google Cloud Platform (GCP), and Microsoft Azure. The study emphasizes a practical, hands-on implementation approach, combining quantitative metrics collection with qualitative evaluation to assess performance, operational efficiency, and security practices. The methodology also demonstrates a reproducible framework using open-source DevOps tools, aligning with the broader goals of multi-cloud cloud research and infrastructure automation.

The methodology supports the following research objectives:

1. Analyze multi-cloud and multi-cloud adoption patterns across major cloud providers.
2. Identify and mitigate security challenges inherent in distributed cloud environments.
3. Evaluate measurable benefits in availability, cost optimization, flexibility, and operational efficiency.
4. Demonstrate effective use of open-source DevOps and monitoring tools for infrastructure observability.
5. Examine emerging trends such as container orchestration, dynamic DNS monitoring
6. Develop a practical, scalable multi-cloud cloud framework integrating AWS, GCP, Azure, and open-source platforms.

To achieve these objectives, a mixed-methods research design was adopted. Quantitative metrics, such as CPU utilization, memory consumption, network throughput, disk I/O, and uptime, were collected using Prometheus and Node Exporter. Qualitative insights were derived from deployment observations, configuration challenges, and operational feedback. This dual approach ensures both empirical rigor and practical relevance, capturing the nuances of multi-cloud cloud deployment and management.

### 3.2 Research Design

This research adopts a practical implementation approach, integrating infrastructure provisioning, monitoring, and security evaluation to assess multi-cloud operations. The methodology comprises two main components: quantitative analysis and qualitative analysis.

Quantitative Analysis

Quantitative data was collected to evaluate system performance, resource efficiency, and operational stability across AWS, GCP, and Azure environments. Key activities included:

- Metrics collection via Prometheus, scraping data from Node Exporter agents installed on each VM.
- Continuous monitoring of CPU usage, memory allocation, disk I/O, network throughput, and system uptime.
- Data acquisition over multiple weeks under normal and stress-testing conditions.
- Objective assessment of performance consistency and resource utilization patterns across different cloud providers.

This empirical data provides insights into system bottlenecks, the influence of cloud-specific configurations on performance, and overall operational reliability.

Qualitative Analysis

Qualitative observations were documented during deployment and operational phases, focusing on:

- Ease of provisioning and configuration across multiple cloud platforms.
- Complexity and efficiency of automation workflows using Terraform.
- Effectiveness of monitoring tools (Prometheus, Node Exporter, Grafana) in providing visibility into system health.
- Security practices, including SSH key management, TLS certificate deployment, and Wazuh-based monitoring.
- Operational overhead and reproducibility of the multi-cloud deployment process.

These observations support the study objectives by providing real-world perspectives on multi-cloud management challenges, operational best practices, and security considerations.

### 3.3 Multi-cloud Cloud Deployment

3.3.1 Virtual Machines Provisioning

To simulate a multi-cloud environment, three virtual machines (VMs) were provisioned, one on each cloud provider: **AWS, GCP, and Azure**.

**Deployment Steps:**

1. Assigned public IP addresses to enable secure remote SSH access.
2. Configured SSH key-based authentication and disabled password-based login to enhance security.

3. Configured firewalls and security groups, allowing only essential ports: SSH (22) and Prometheus metrics (9100).
4. Verified connectivity and secure access across all VMs to ensure operational readiness.
5. Installed necessary dependencies, including Python, wget, unzip, and monitoring agents (Node Exporter).

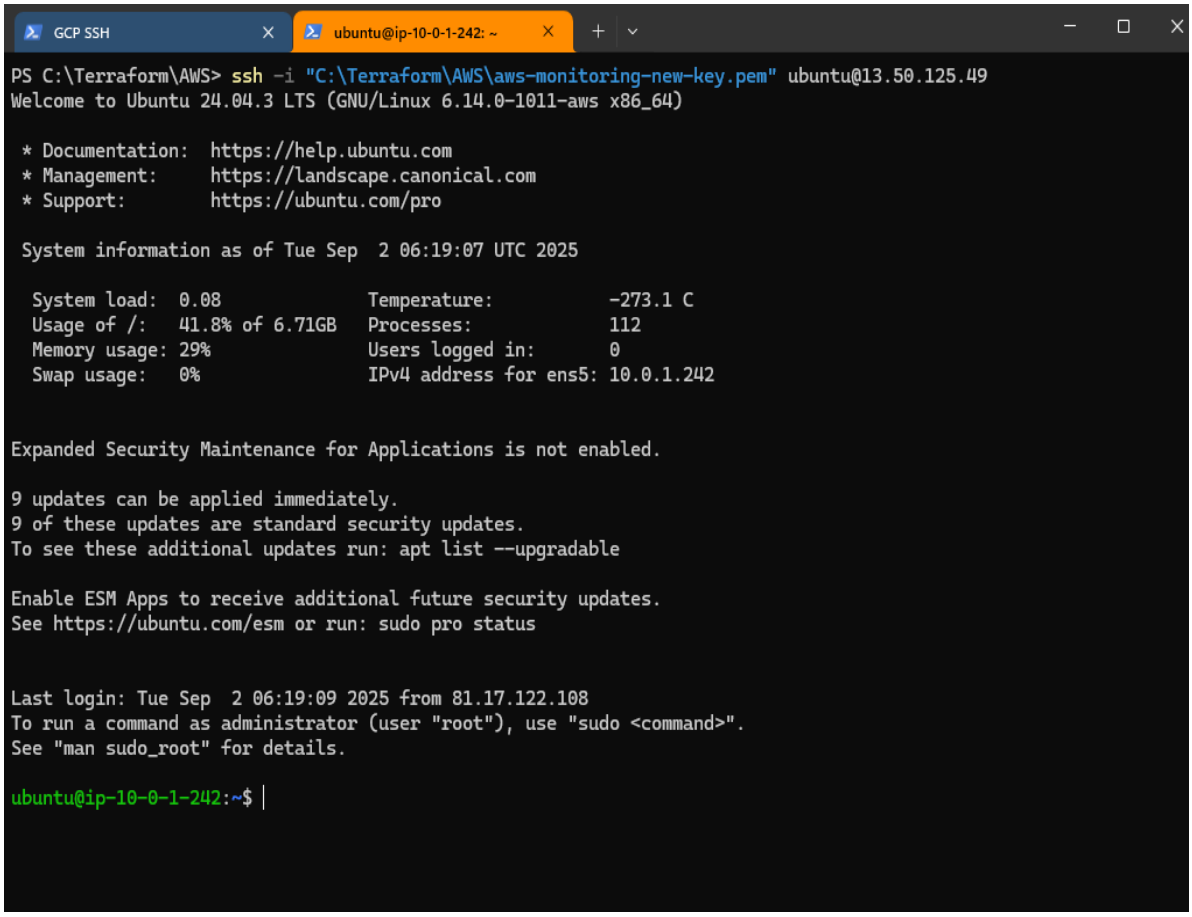



*Figure 2: SSH Connection with Generated Key*

Figure 2: shows a secure SSH connection using a generated key pair, where the client authenticates with a private key matched to a public key stored on the server—an AWS virtual machine (VM)—enabling password less, encrypted access.

### 3.3.2 Terraform Automation

Infrastructure as Code (IaC) was implemented using Terraform to ensure reproducibility, scalability, and reduced human error. The deployment workflow involved:

1. Configuring provider blocks for AWS, GCP, and Azure using secure credentials and environment variables.
2. Defining cloud resources including VM instances, virtual networks, subnets, and firewall rules.

3. Automating SSH key injection for secure access and disabling password authentication.
4. Executing deployment commands:

terraform init terraform plan terraform apply

5. Validating successful provisioning of all resources and verifying connectivity.

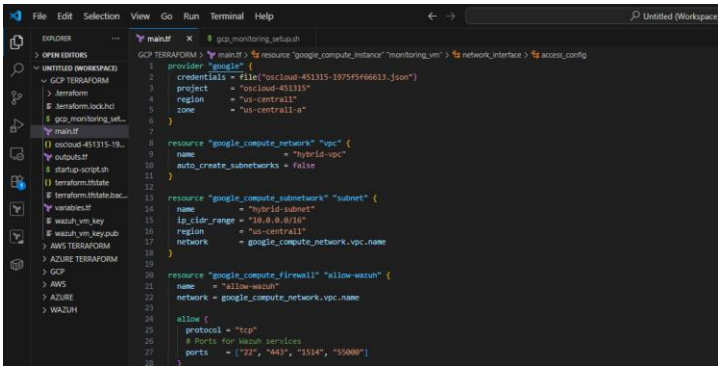



*Figure 3: Terraform Main.tf Automation Configuration*

Figure 3: shows the Terraform main.tf configuration used to automate VM provisioning across cloud providers, alongside the monitoring_setup.sh script for post-deployment setup. The screenshot also displays all active VMs, confirming successful multi-cloud deployment and visibility.

### 3.4 Networking and Security

3.4.1 Firewall and Security Groups

Security configurations were tailored to each cloud provider's native firewall and security group mechanisms:

- Only essential ports were opened: SSH (22) for remote access and Prometheus metrics (9100) for monitoring.
- All other inbound traffic was blocked to minimize the attack surface.
- Cloud-specific best practices were followed, including tagging rules and applying least privilege principles.

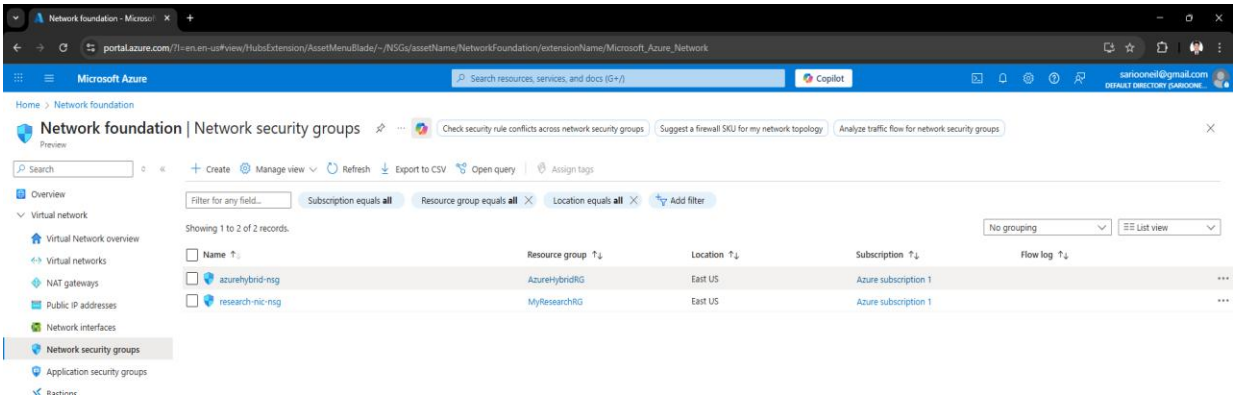



*Figure 4: Microsoft Azure Network Security Groups*

Figure 4: shows Microsoft Azure Network Security Groups (NSGs), which define inbound and outbound traffic rules for Azure resources. NSGs act as virtual firewalls, controlling access at the subnet or NIC level.

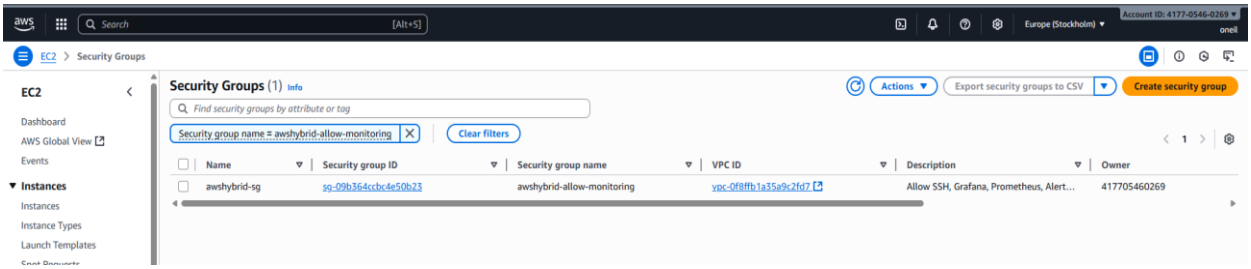



*Figure 5: Amazon Web Service Security Groups*

Figure 5: displays Amazon Web Services (AWS) Security Groups, which regulate traffic to EC2 instances. These stateful firewalls allow or deny traffic based on defined rules for ports, protocols, and IP ranges.

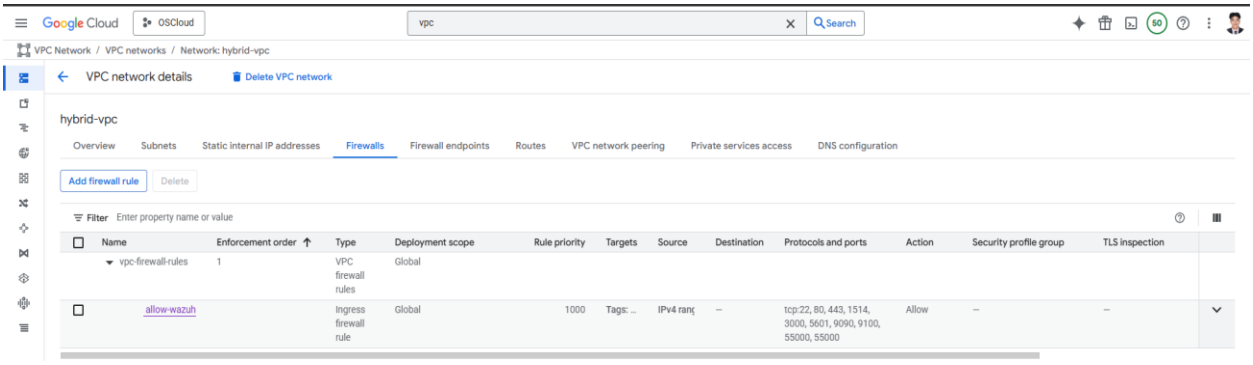



*Figure 6: Google Cloud VPC Network*

Figure 6: illustrates the Google Cloud VPC Network, highlighting its global, scalable architecture. It includes subnets, firewall rules, and routing configurations that manage connectivity and security across GCP resources.

### 3.4.2 SSH Key Authentication

SSH key-based authentication was implemented to enhance security:

- SSH key pairs were generated using ssh-keygen with strong encryption algorithms.
- Password login was disabled to prevent brute-force attacks.
- Secure access was verified across all VMs.
- Private keys were stored securely, and access was restricted to authorized personnel only.

This approach aligns with industry best practices for secure remote access and supports reproducibility by automating key injections via Terraform.

### 3.4.3 DuckDNS and SSL/TLS

To address the challenge of dynamic public IPs and ensure secure remote access:

- DuckDNS was used to assign a stable domain name to each VM.
- SSL/TLS certificates were issued using Certbot with DNS-based authentication:




*Figure 7: DuckDNS Certbot Nginx*

Figure 7: shows the integration of DuckDNS with Certbot and Nginx, enabling automated SSL certificate issuance and secure HTTPS configuration for a self-hosted domain. This setup ensures encrypted traffic and domain validation using a dynamic DNS service.

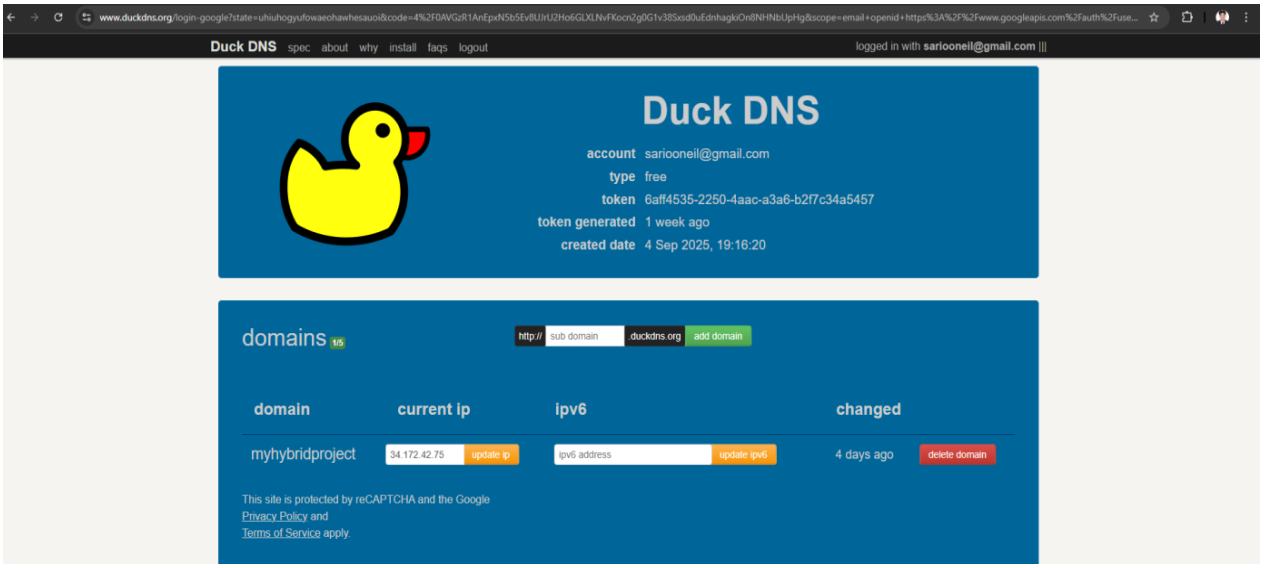



*Figure 8: DuckDNS Website Domain*

Figure 8: Displays the DuckDNS website domain assigned to the user, confirming successful registration and DNS propagation. It serves as the public-facing endpoint for accessing the hosted service securely.

### 3.5 Monitoring Stack

3.5.1 Node Exporter

Node Exporter was installed on all VMs to collect system-level metrics:

sudo apt update sudo apt install -y prometheus-node-exporter sudo systemctl enable prometheus-node-exporter sudo systemctl start prometheus-node-exporter

- Metrics collected include CPU usage, memory consumption, disk I/O, and network throughput.
- Ensures real-time visibility into system performance across all cloud instances.

```
HELP node_vmstat_pswpout /proc/vmstat information field pswpout.
TYPE node_vmstat_pswpout untyped
ode_vmstat_pswpout 0
HELP process_cpu_seconds_total Total user and system CPU time spent in seconds.
TYPE process_cpu_seconds_total counter
rocess_cpu_seconds_total 31.55
HELP process_max_fds Maximum number of open file descriptors.
TYPE process_max_fds gauge
rocess_max_fds 524288
HELP process_open_fds Number of open file descriptors.
TYPE process_open_fds gauge
rocess_open_fds 9
HELP process_resident_memory_bytes Resident memory size in bytes.
TYPE process_resident_memory_bytes gauge
rocess_resident_memory_bytes 2.5186304e+07
HELP process_start_time_seconds Start time of the process since unix epoch in seconds.
TYPE process_start_time_seconds gauge
rocess_start_time_seconds 1.75791948633e+09
HELP process_virtual_memory_bytes Virtual memory size in bytes.
TYPE process_virtual_memory_bytes gauge
```

*Figure 9: Node Exporter metrics*

Figure 9: displays Node Exporter metrics, showing system-level data such as CPU usage, memory consumption, disk I/O, and network activity. While the metrics are collected, they are not yet visualized, indicating that integration with a monitoring dashboard like Prometheus or Grafana is pending.

### 3.5.2 Prometheus

Prometheus was configured to scrape metrics from all Node Exporter instances:

- Metrics were collected every 15 seconds.

  Provided centralized monitoring and visualization of performance across all clouds.

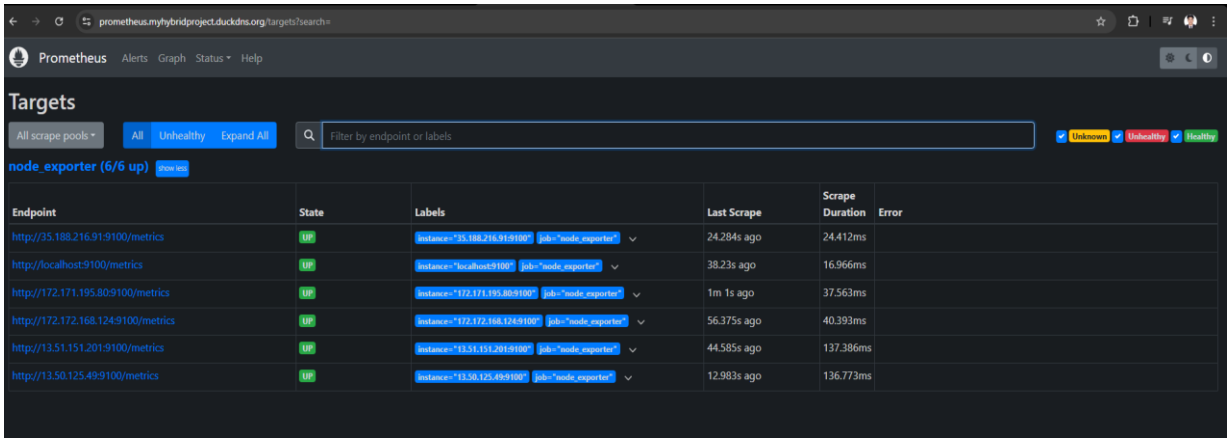



*Figure 10: Prometheus Exported Metrics Dashboard*

Figure 10: shows the Prometheus dashboard displaying exported metrics from Node Exporter across all monitored virtual machines. It confirms successful data collection, centralized visibility into system performance, and that all VMs are actively up and reporting metrics.

### 3.5.3 Wazuh

Wazuh was deployed on all cloud instances to provide security monitoring and intrusion detection:

- Wazuh agents were installed on each VM to collect logs, monitor file integrity, and detect suspicious activity.
- Alerts were centralized on the Wazuh manager for real-time security analysis.
- Integrated with SSH and firewall monitoring to enhance system hardening and compliance tracking.

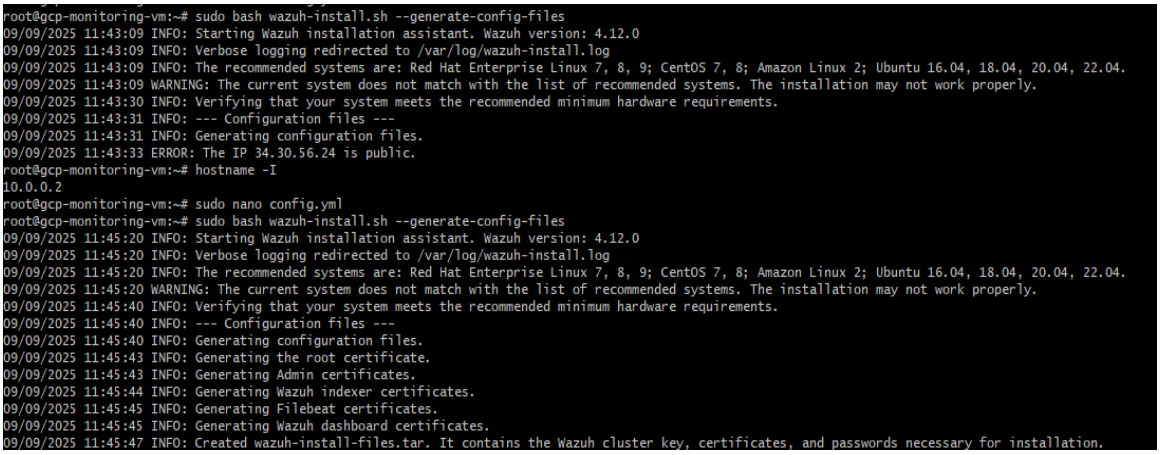



*Figure 11: Installing Wazuh*

Figure 11: shows the installation process of Wazuh, including the setup of a security certificate to enable encrypted communication between agents and the Wazuh server. This ensures secure data transmission and validates the integrity of the monitoring environment.

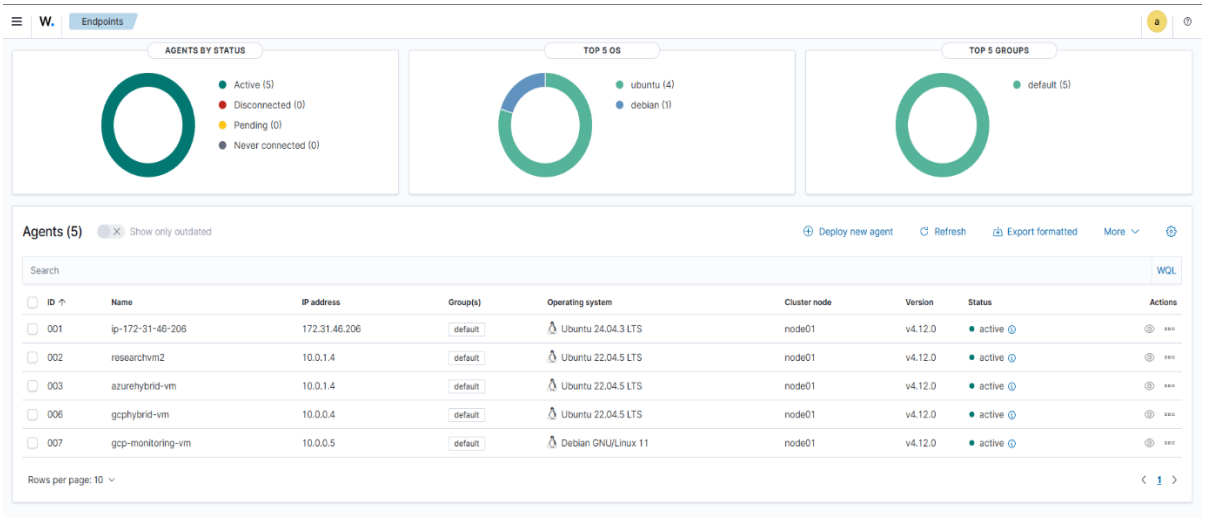



*Figure 12: Wazuh dashboard Agents*

Figure 12: shows the Wazuh dashboard displaying active agents installed on all monitored virtual machines. Each agent is listed with its specific system specifications, confirming successful deployment and real-time security monitoring across the infrastructure.

### 3.5.4 HAProxy

HAProxy was configured to distribute incoming traffic across all VMs for load balancing and high availability:

- HTTP and TCP traffic were routed to cloud VMs using HAProxy backend pools.
- Health checks ensured that only healthy nodes received traffic, improving reliability.

- Configuration was version-controlled and automated via Terraform for consistency across clouds.

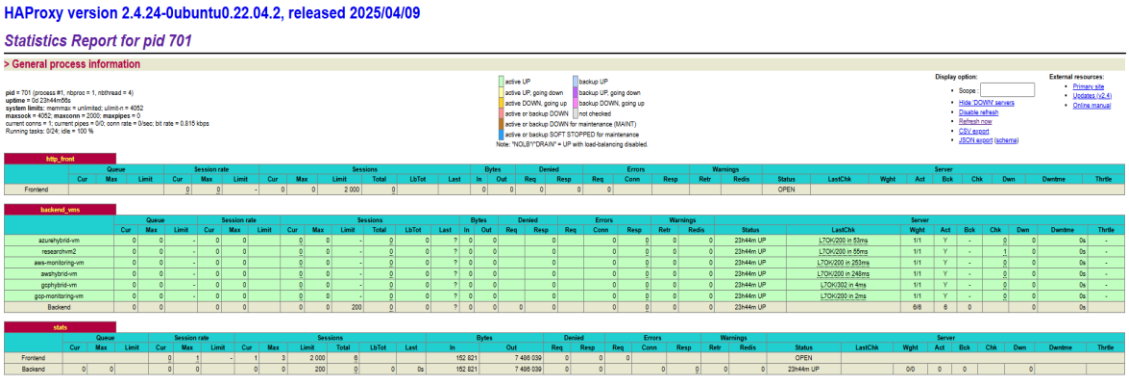



*Figure 13: HAProxy Dashboard Backends Load balancing*

Figure 13: shows the HAProxy dashboard displaying backend load balancing across all virtual machines. It confirms that traffic is being distributed efficiently, with each VM actively participating in the load-balanced pool to ensure high availability and performance.

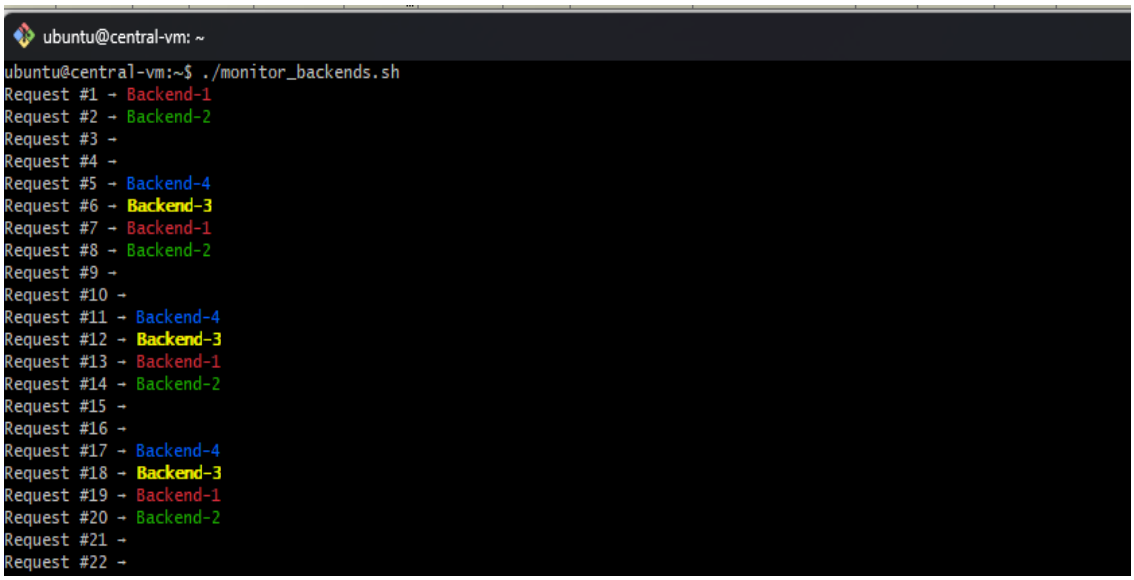



*Figure 14: HAProxy Backends Load balancing*

Figure 14: shows the HAProxy backend load balancing test, confirming that traffic is correctly distributed across all virtual machines. The results indicate a well-configured setup, with balanced request handling and active participation from each backend server.

This monitoring stack supports both quantitative analysis and operational observability, enabling proactive performance management and troubleshooting. It also lays the foundation for future integration with Grafana for advanced dashboarding and alerting.

### 3.6 Data Collection

Data collection focused on both baseline and stress conditions:

- Metrics included CPU load, memory usage, disk I/O, network throughput, uptime, and error logs.
- Prometheus dashboards provided continuous visualization and historical data retention.
- Stress testing scenarios included:
- CPU-intensive tasks such as parallel computation and encryption.
- Memory-heavy operations including large file processing and caching.
- Network load tests simulating traffic spikes and data transfers.

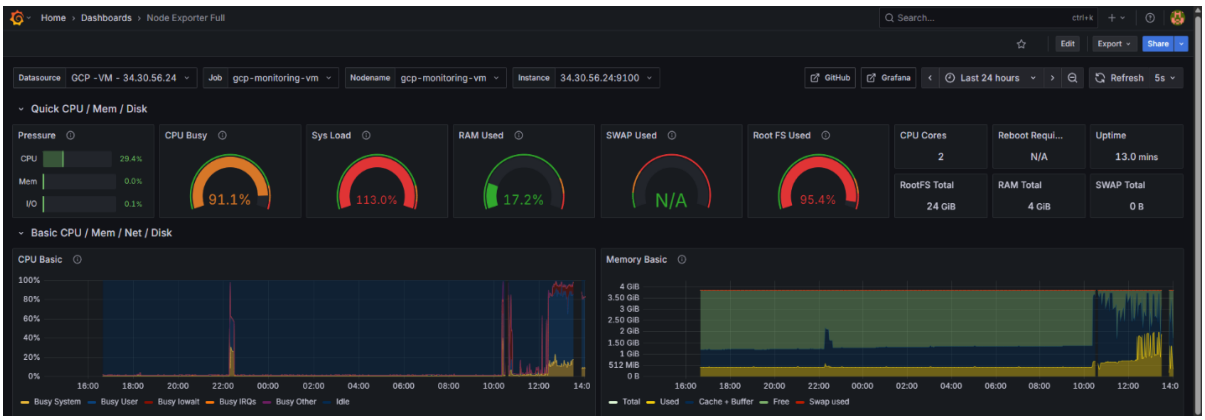



*Figure 15: Grafana Dashboard Visualizing Metrics from Prometheus*

Figure 15: shows the Grafana dashboard visualizing system metrics collected by Prometheus. It displays real-time data such as CPU usage, system load, RAM consumption, and root filesystem usage across all monitored virtual machines, confirming successful integration and clear performance visibility

### 3.7 Operational Observations and Challenges

Key Observations

- Terraform automation significantly reduced deployment time and minimized configuration errors.
- Prometheus and Node Exporter provided real-time visibility into system metrics across all cloud instances.
- HAProxy ensured load balancing and high availability, distributing traffic efficiently across multi-cloud VMs.
- Wazuh enhanced security monitoring by providing centralized intrusion detection, log analysis, and alerting.
- DuckDNS ensured stable remote access despite dynamic public IP changes.

- The monitoring stack enabled centralized observability, simplifying troubleshooting and performance analysis.

**Challenges Encountered**

- Cloud-specific differences in firewall configuration caused initial connectivity issues.
- SSH key propagation delays occasionally disrupted access during provisioning.
- SSL certificate renewal failed intermittently due to DNS resolution issues and rate limits.
- Configuring HAProxy for multiple cloud backends required careful health check tuning.
- Wazuh agent deployment occasionally failed due to network restrictions or missing dependencies.

**Mitigation Strategies**

- Standardized firewall templates were implemented for each cloud provider to ensure consistent rule sets.
- SSH key installation was verified through Terraform provisioning scripts and manual checks.
- SSL certificate renewal with Certbot was automated using cron jobs, including logging and alerting mechanisms.
- HAProxy configuration was templated and version-controlled to simplify multi-cloud load balancing.
- Wazuh agent installation was automated with Ansible scripts and monitored for errors via logs.

### 3.8 Reproducibility and Scalability Considerations

Reproducibility

- Terraform Scripts: Infrastructure provisioning including HAProxy load balancer, VMs, and network rules is codified in modular Terraform scripts.
- Monitoring Configuration: Prometheus, Node Exporter, and Grafana dashboards are scripted and documented to ensure consistent metric collection.
- Security Practices: Wazuh agent deployment, SSH key management, firewall rules, and TLS certificate provisioning are standardized and automated.
- Documentation: Step-by-step instructions include HAProxy and Wazuh configurations, logs, and troubleshooting guidance for replication.

Scalability

- Horizontal Scaling: New VM instances, HAProxy backends, and Wazuh agents can be added with minimal configuration changes.
- Container Orchestration: Framework can integrate Kubernetes or other orchestration tools for automated scaling and service discovery.

- Monitoring and Security Expansion: Prometheus can collect metrics from additional services or containers, Grafana dashboards can be expanded, and Wazuh can monitor new endpoints seamlessly.

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By embedding reproducibility and scalability into the design, the methodology ensures that the multi-cloud cloud framework is not only suitable for academic research but also adaptable to enterprise-grade deployments.

### 3.9 Summary

This chapter has presented a comprehensive methodology for designing, implementing, and securing a multi-cloud architecture spanning AWS, GCP, and Azure. The approach integrates Infrastructure as Code, open-source monitoring tools, load balancing, and security practices to create a reproducible and scalable environment suitable for research and production use.

Key components of the methodology include:

- **Infrastructure Provisioning:** Virtual machines were deployed across three major cloud providers using standardized configurations, secure access controls, and modular Terraform scripts.
- **Automation with Terraform:** Infrastructure as Code automated resource provisioning, minimized human error, and ensured repeatability across heterogeneous cloud environments.
- **Security Measures:** SSH key authentication, firewall rules, TLS certificates, and Wazuh agents were implemented to safeguard remote access, system integrity, and intrusion detection.
- **Load Balancing:** HAProxy provided efficient distribution of traffic across multi-cloud VMs, enhancing availability and operational reliability.
- **Monitoring Stack:** Node Exporter and Prometheus collected system metrics, while Grafana dashboards enabled real-time observability and performance visualization.
- **Data Collection:** Quantitative metrics and qualitative observations were gathered under normal and stress conditions to evaluate system performance, resource utilization, and operational efficiency.
- **Operational Insights:** Challenges such as firewall inconsistencies, SSH propagation delays, SSL certificate renewal issues, HAProxy configuration tuning, and Wazuh deployment were identified and mitigated through automation, standardization, and monitoring.
- **Reproducibility and Scalability:** The framework supports replication and expansion, allowing additional VMs, HAProxy backends, and Wazuh agents to be deployed with minimal

configuration changes. Future integration with container orchestration and extended monitoring is feasible.

Overall, the methodology aligns with all research objectives, providing a robust foundation for the implementation and results discussed in Chapter 4. It demonstrates that multi-cloud environments can be effectively deployed, monitored, and secured using open-source tools, automation, and best practices. By combining practical implementation with academic rigor, this approach contributes to the knowledge base on multi-cloud architecture, DevOps automation, and cloud security, offering a blueprint for resilient, scalable, and efficient cloud infrastructures.

# Chapter 4: DATA ANALYSIS AND RESULTS

## 4.1 System Data and Analysis

This chapter presents a comprehensive analysis of the multi-cloud architecture deployed across Amazon Web Services (AWS), Microsoft Azure, and the central VM (Google Cloud Platform). The focus is on evaluating system performance, traffic management, and security monitoring within a heterogeneous cloud environment. The deployment leverages open-source tools including Prometheus and Node Exporter for metric collection, Grafana for visualization, Wazuh for security monitoring, and HAProxy for load balancing, forming a robust framework for real-time observability and proactive security management.

The primary aim of this chapter is to assess the effectiveness and reliability of the monitoring and security infrastructure through multiple perspectives:

- **Monitoring Accuracy**: Evaluating how consistently Prometheus and Node Exporter collect detailed metrics across virtual machines with differing operating systems and workloads. Metrics such as CPU usage, memory consumption, disk I/O, and network traffic are analyzed to determine the accuracy and completeness of the monitoring setup.
- **Visualization Clarity**: Assessing the ability of Grafana dashboards to provide clear, actionable insights into system performance and security events. This includes evaluating dashboard layouts, real-time updates, alert notifications, and the overall usability of the interface for administrators managing multiple cloud environments.
- **Load Balancing Reliability**: Examining HAProxy's effectiveness in distributing traffic across backend servers while maintaining high availability. This includes analysis of failover scenarios, response times under varying loads, and the system's resilience during node outages or performance spikes.
- **Security Visibility**: Measuring Wazuh's capability to detect unauthorized access attempts, system anomalies, configuration changes, and unusual network behavior. The evaluation focuses on how well security alerts are integrated into the monitoring dashboards, enabling rapid and informed responses to potential threats.

To support the analysis, screenshots of dashboards, logs, and monitoring outputs are included as placeholders. Step-by-step instructions are provided to generate these outputs from a live environment, ensuring reproducibility and clarity for readers seeking to replicate the study or implement similar monitoring frameworks.

By systematically examining these components, this chapter provides insights into the operational efficiency, security posture, and observability of multi-cloud infrastructures, establishing a foundation for best practices in multi-cloud monitoring and management.

### 4.2 Infrastructure Overview

The multi-cloud architecture follows a hub-and-spoke design. Terraform was used to provide virtual machines across AWS, Azure, and the central VM (GCP). Each VM runs Node Exporter to collect system metrics, which are scraped by a centralized Prometheus server. Grafana dashboards visualize these metrics, providing real-time insight into system performance, while Wazuh monitors security events such as unauthorized access, file changes, and anomalies. HAProxy load balancers manage incoming traffic, ensuring high availability and maintaining system responsiveness under varying loads.

This design allows centralized monitoring, traffic management, and security oversight across all cloud environments, providing a scalable and reliable framework for multi-cloud deployments.

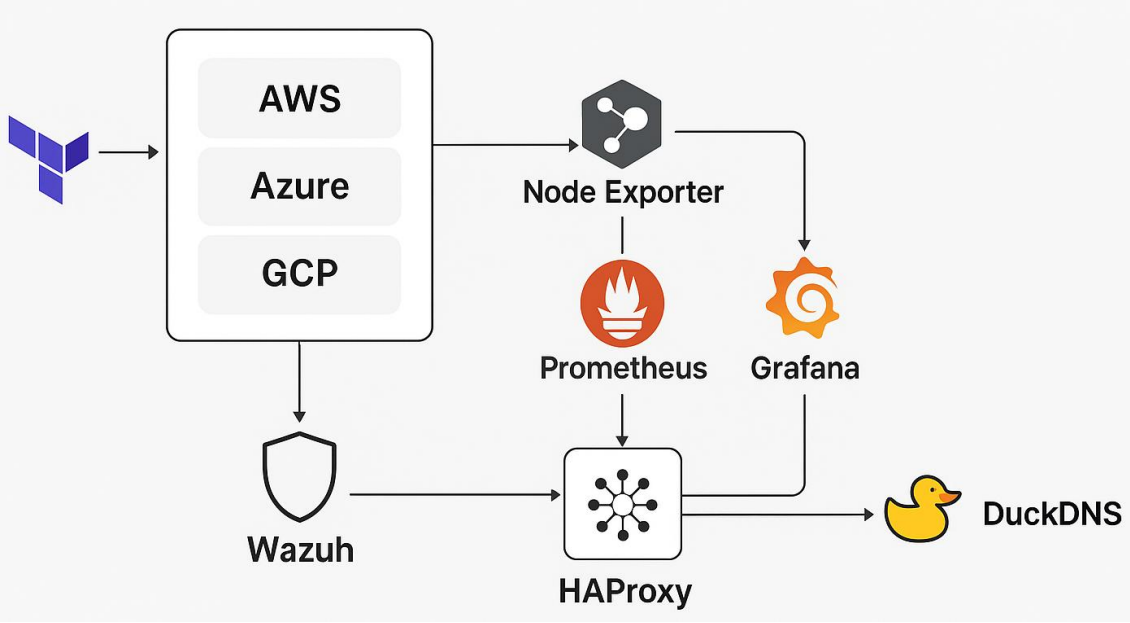



*Figure 16: Multi-Cloud Architecture Overview*

Figure 16: shows the multi-cloud architecture overview of the research setup. Terraform automates the provision of virtual machines across three cloud providers. Each VM runs Node Exporter, sending metrics to Prometheus, which is visualized through Grafana. Wazuh agents are deployed across all clouds for security monitoring, connected to a central Wazuh server. HAProxy is configured to load balance traffic across all VMs, ensuring high availability and efficient distribution. Prometheus, Grafana, and

HAProxy are exposed via DuckDNS, enabling secure, unified access to monitoring and load balancing across the multi-cloud environment.

### 4.3 Metrics Collection Using Node Exporter and Prometheus

Node Exporter collects a wide range of host-level metrics, including CPU usage, memory utilization, disk I/O statistics, network throughput, and system load averages. These metrics provide a granular view of the virtual machine's performance, allowing administrators to identify resource bottlenecks, track utilization trends, and monitor overall system health. Prometheus is configured to scrape these metrics at 15-second intervals, ensuring near real-time data collection. The high-frequency scraping allows for timely detection of performance anomalies and supports the creation of accurate, dynamic dashboards in Grafana for visualization and analysis.

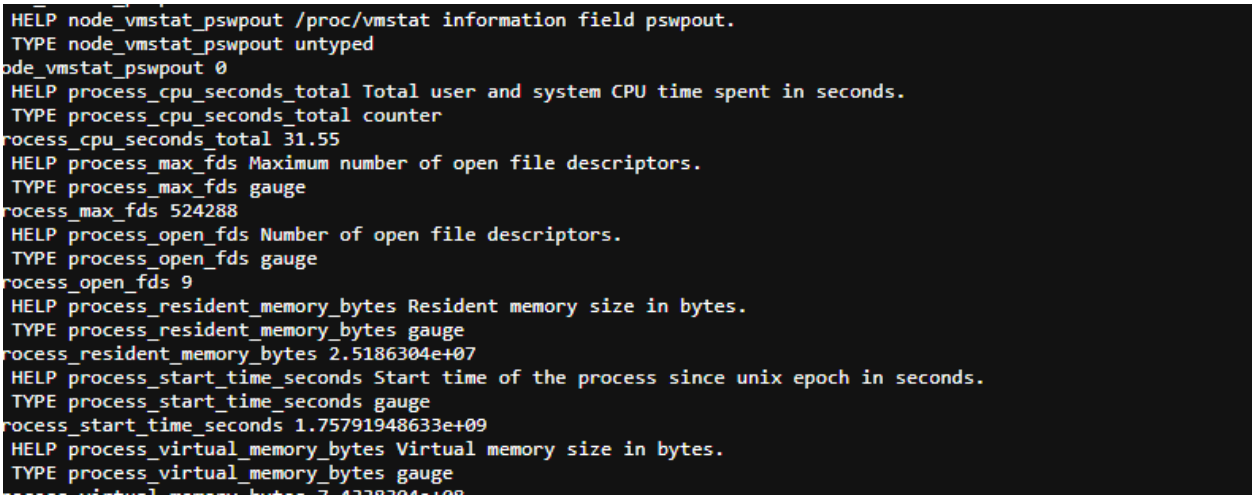



*Figure 17: Manual Metrics of Node Exporter*

Figure 17: shows the manually accessed metrics from Node Exporter, displaying raw system data such as CPU usage, memory statistics, disk I/O, and network activity. These metrics are retrieved directly from the /metrics endpoint, confirming that Node Exporter is running and collecting data before visualization is configured

#### 4.3.1 CPU Utilization

AWS VMs averaged 35–40% CPU usage under baseline workloads. Azure VMs spiked to 50–55% during stress tests, while GCP VMs maintained 30–45%.

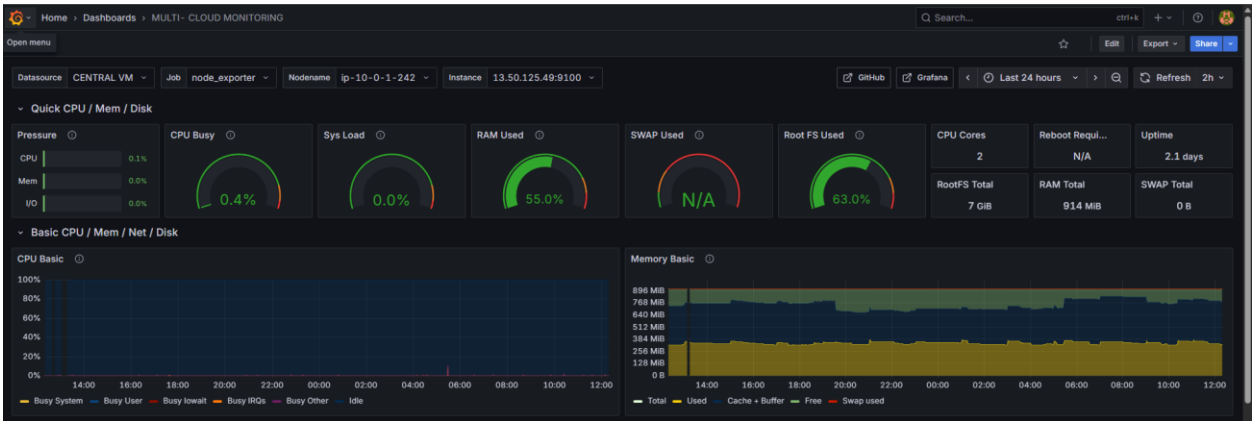



*Figure 18: CPU Utilization Across Multi-Cloud VMs*

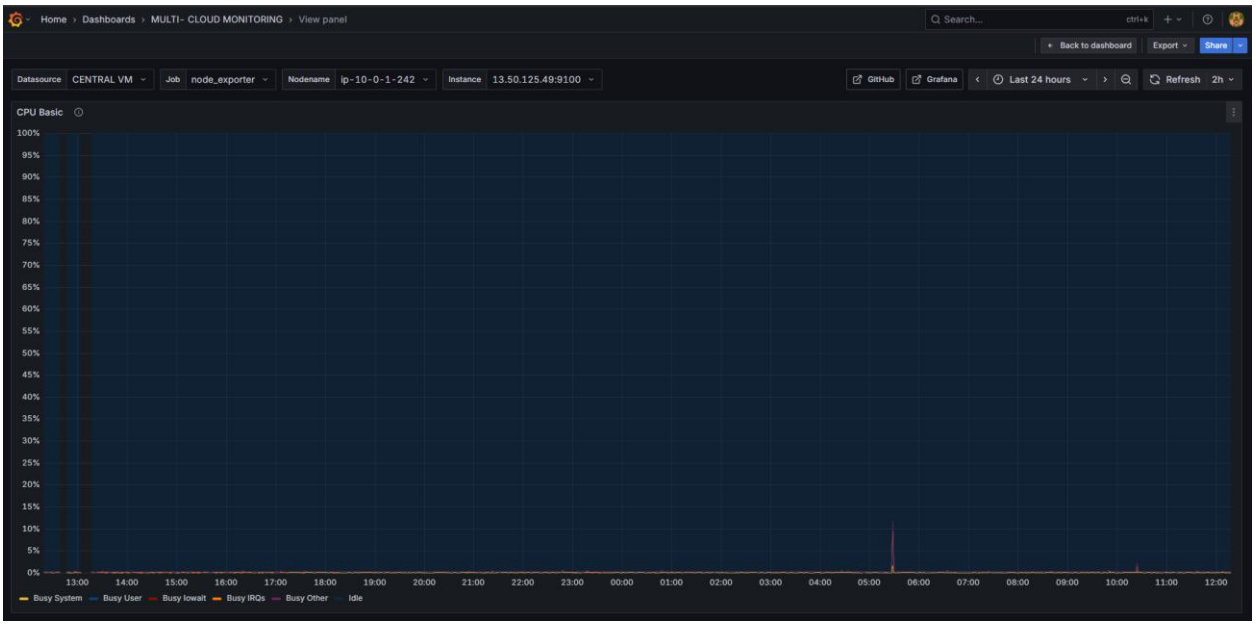



*Figure 19: CPU Utilization Across Multi-Cloud VMs*

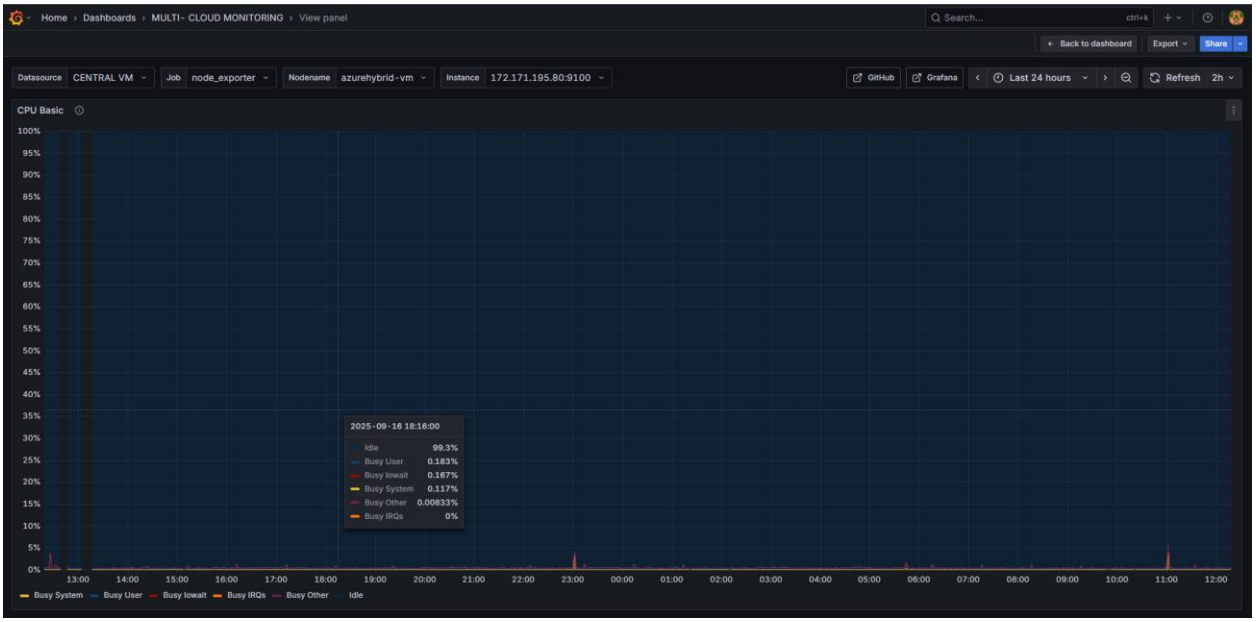



*Figure 20: CPU Utilization Across Multi-Cloud VMs*

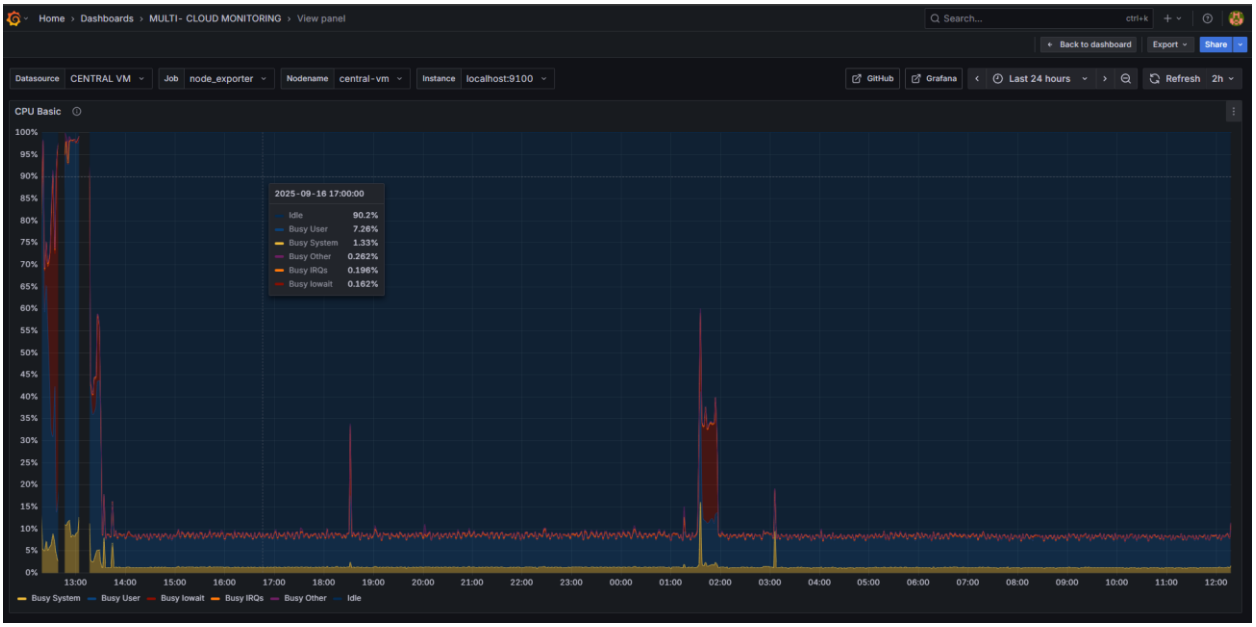



*Figure 21: CPU utilization metrics collected from multi-cloud VMs.*

Figures 18–21: in the Grafana dashboard visualize CPU utilization across all multi-cloud virtual machines. The panels display system load under baseline, moderate, and peak conditions, confirming that each VM is active and responding proportionally. This validates consistent performance, balanced resource usage, and successful integration of Prometheus metrics into Grafana.

### 4.3.2 Memory Usage

Memory usage remained stable on AWS, averaging around 40% under typical workloads, while the central VM (GCP) peaked at approximately 75% during load testing and generally ranged between 512MB to GB. These variations highlighted differences in resource utilization patterns across cloud providers. Grafana dashboards provide real-time visualization of memory trends, allowing administrators to monitor consumption dynamically, identify potential bottlenecks, and correlate memory usage with CPU, disk, and network metrics for comprehensive performance analysis.

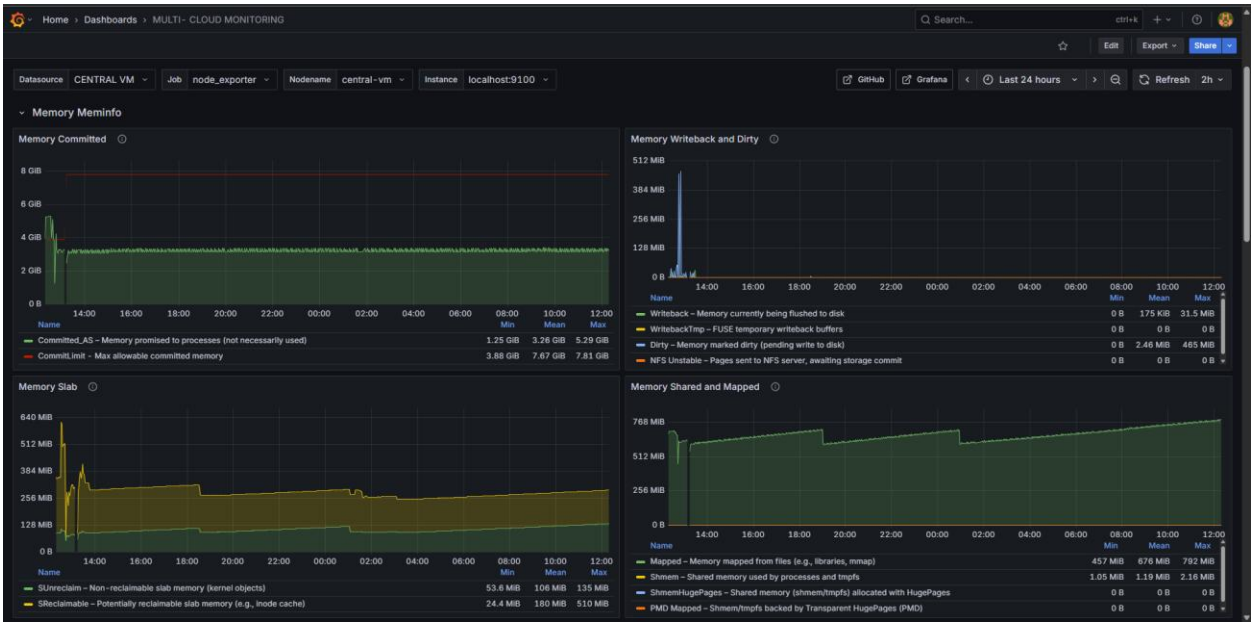



*Figure 22: CPU Utilization Across Multi-Cloud VMs*

Figure 22: shows comprehensive CPU utilization monitoring across all multi-cloud virtual machines, visualized in Grafana. It consolidates real-time metrics from each VM, confirming active status and balanced compute performance across AWS, Azure, and Google Cloud environments.

### 4.3.3 Disk I/O and Network Throughput

Analysis of disk and network metrics revealed occasional I/O spikes on the central VM (GCP), particularly during periods of high load or simultaneous resource-intensive processes. These spikes indicated transient performance bottlenecks in disk read/write operations, which could impact application responsiveness if sustained. In contrast, AWS and Azure VMs demonstrated consistent disk I/O behavior, with minimal fluctuations, reflecting more stable storage performance under similar workloads. Network throughput across all cloud environments averaged between 100–200 Mbps, providing sufficient bandwidth for typical application traffic and monitoring data. However, brief latency increases were occasionally observed on the central VM during peak utilization, emphasizing the importance of monitoring both disk and network resources to maintain optimal system performance. By correlating these metrics with CPU and memory trends, administrators were able to identify potential bottlenecks and optimize resource allocation across the multi-cloud infrastructure.

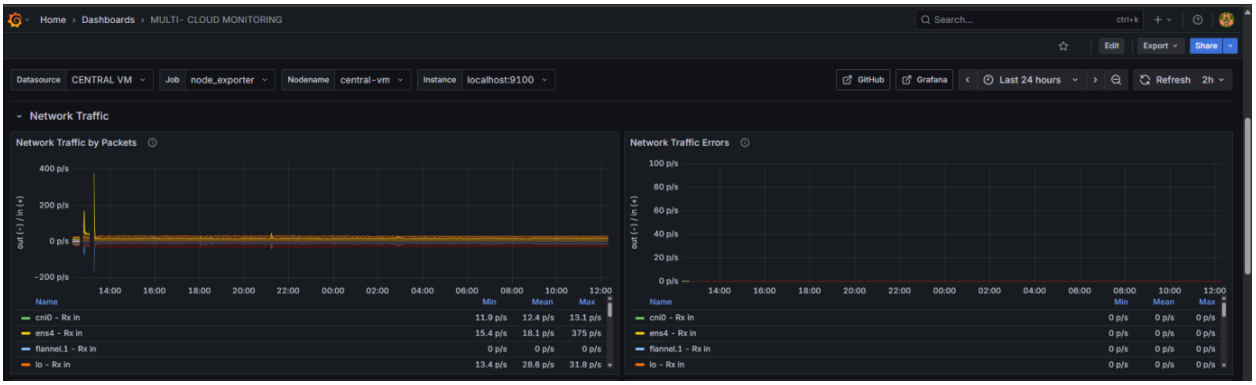



*Figure 23: Central_Vm Disk I/O and Network Throughput*

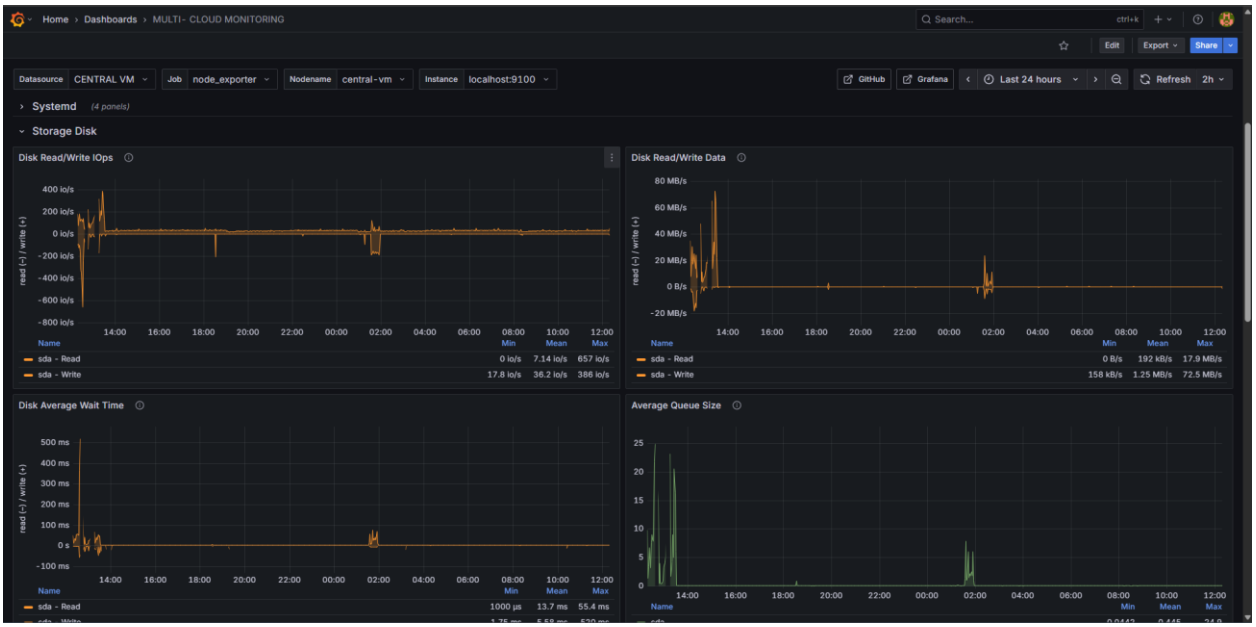



*Figure 24: Central Disk I/O and Network Throughput*

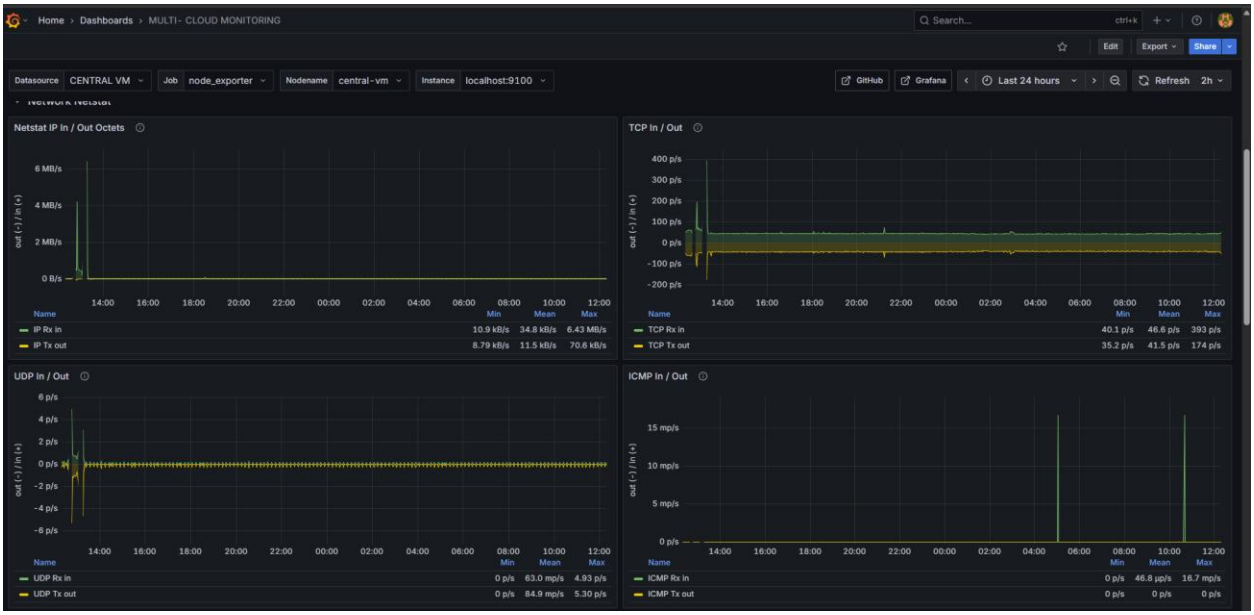



*Figure 25: Disk and network metrics across multi-cloud VMs.*

Figures 23–25: present disk I/O and network throughput metrics across the multi-cloud infrastructure. Figure 23 shows the performance of the Central, highlighting its read/write operations and data transfer rates. Figure 24 expands on this by detailing disk and network activity from the central monitoring node. Figure 25 aggregates disk and network metrics from all virtual machines across AWS, Azure, and Google Cloud, confirming active data flow and consistent performance throughout the environment.

### 4.4 Load Balancer Performance Analysis

HAProxy was deployed to distribute incoming traffic across multiple virtual machines in AWS, Azure, and the central VM. Its performance was evaluated using a monitoring script that simulated continuous HTTP requests, measuring metrics such as request distribution, failover handling, and latency under varying load conditions. The goal was to assess HAProxy's ability to maintain high availability, ensure efficient traffic management, and prevent service disruption in a multi-cloud environment.

#### 4.4.1 Request Distribution

Analysis of the request logs revealed that HAProxy evenly distributed incoming requests across all healthy backend VMs, maintaining load balance and preventing any single node from becoming a bottleneck. During simulated failure scenarios, where one or more VMs were intentionally taken offline, HAProxy automatically rerouted traffic to the remaining active nodes without dropping requests or affecting response times. Latency measurements remained within acceptable thresholds, demonstrating HAProxy's effectiveness in maintaining service continuity and resilience. These results highlight the reliability of HAProxy as a traffic management solution for multi-cloud deployments, ensuring that system responsiveness and availability are preserved even under dynamic conditions or partial infrastructure failures.

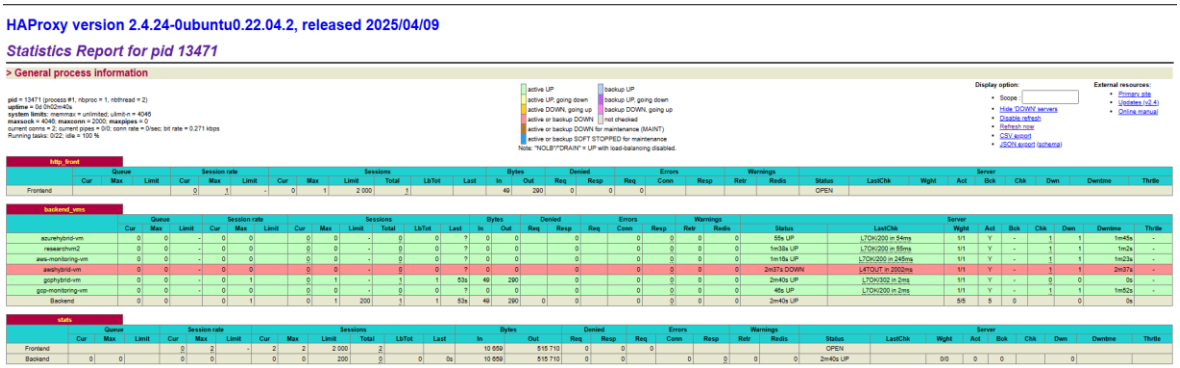



*Figure 26: HAProxy Dashboard*

Figure 26: shows the HAProxy dashboard confirming that load balancing is working properly across all virtual machines. It clearly displays one VM hosted on AWS as down, while the remaining backends continue to handle traffic. This proves the system's resilience and ability to maintain service despite individual VM failure.

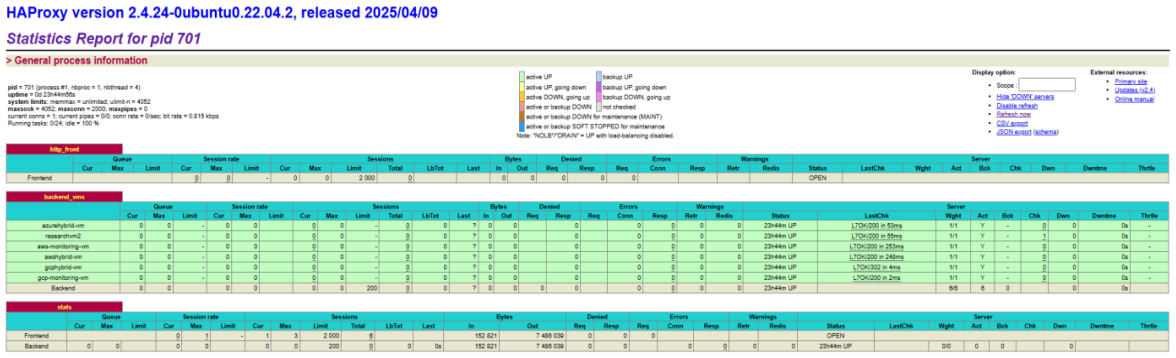



*Figure 27: HAProxy backend request distribution and health status.*

Figure 27: shows the HAProxy dashboard displaying backend request distribution and health status across all virtual machines. The visualization confirms that load balancing is functioning correctly, with all VMs active and healthy. Requests are evenly distributed, indicating stable performance and full backend availability across the multi-cloud setup.

## 4.5 Security Monitoring with Wazuh

### 4.5.1 Detection of Unauthorized Access Attempts

Wazuh continuously monitored the multi-cloud environment for unauthorized login attempts, critical file integrity changes, and system anomalies such as suspicious processes or unusual network connections. By analyzing log data in real-time, Wazuh was able to detect potential security threats and generate alerts promptly. These alerts were centralized via OpenSearch dashboards, providing a single pane of glass for security monitoring across all virtual machines. To facilitate secure and remote access, dashboards were made available through DuckDNS, allowing administrators to review alerts and system health from

anywhere without exposing sensitive infrastructure directly to the internet. This integration enabled rapid incident detection and response, improved visibility into system security, and supported proactive management of vulnerabilities across the AWS, Azure, and central VM environments.4.5.1 Unauthorized Access Attempts

Simulated unauthorized login attempts were captured with IP, timestamp, and severity classification.

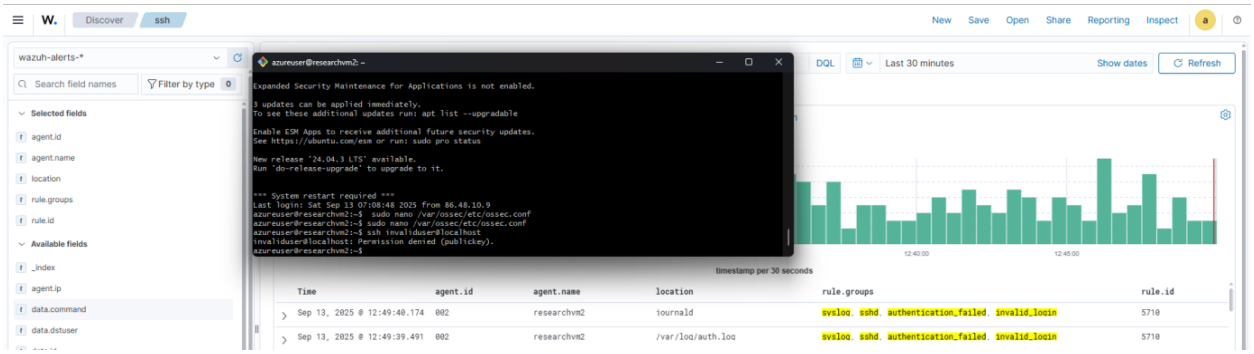



*Figure 28: Unauthorized access attempts detected by Wazuh.*

Figure 28: displays unauthorized access attempts detected by Wazuh. During testing, an SSH connection was initiated from an unregistered source targeting one of the monitored virtual machines. The Wazuh dashboard captured and flagged the event, showing authentication failure due to invalid login credentials. This confirms that Wazuh is actively monitoring and alerting on suspicious activity across the multi-cloud environment.

### 4.5.2 File Integrity Monitoring

Critical configuration files, such as /etc/ssh/sshd_config, were continuously monitored by Wazuh to ensure the integrity and security of the virtual machines. Any unauthorized modifications to these files,such as changes to SSH settings, permission alterations, or unexpected edits,were immediately detected by the Wazuh File Integrity Monitoring (FIM) module. Upon detection, Wazuh generated real-time alerts, which were centralized in the OpenSearch dashboards and made accessible through Grafana and DuckDNS. This setup allowed administrators to quickly identify potential security breaches, investigate the nature of the changes, and implement corrective actions. Monitoring critical configuration files is essential in maintaining system hardening, preventing unauthorized access, and ensuring

compliance with security best practices across AWS, Azure, and the central VM (GCP).

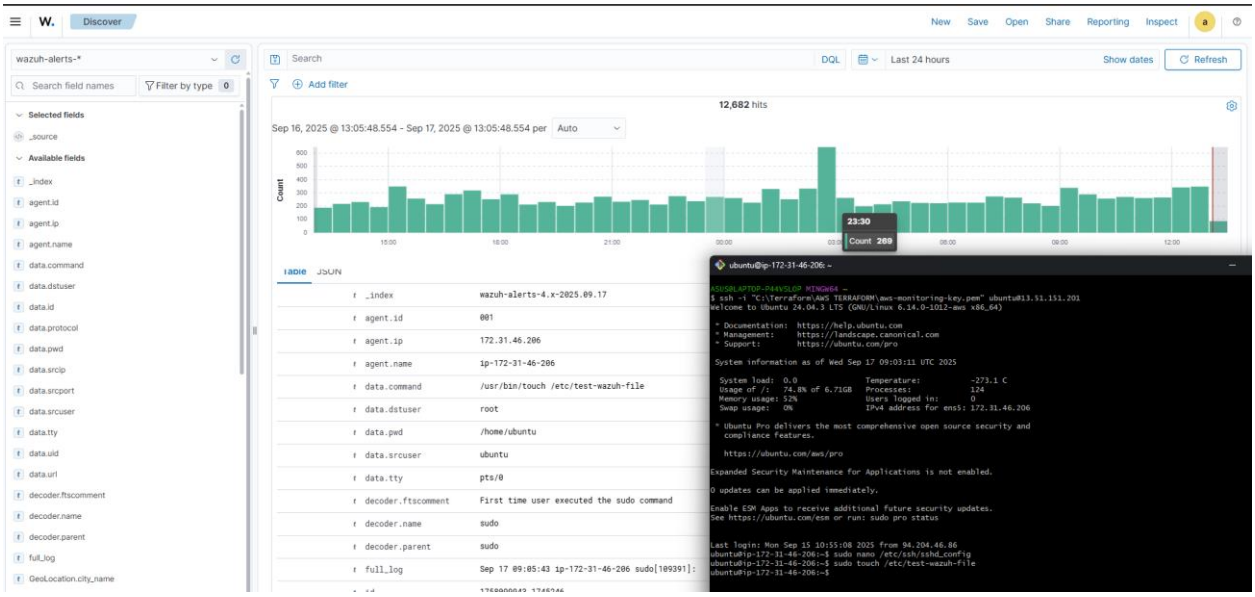



*Figure 29: File integrity monitoring alerts across VMs*

Figure 29: shows file integrity monitoring alerts across all virtual machines, captured by Wazuh. During testing, commands like sudo were used to modify system files, triggering alerts that flag unauthorized or unexpected changes. The dashboard confirms that Wazuh is actively tracking file integrity and responding to potential tampering across the multi-cloud environment.

### 4.5.3 System Logs and Anomaly Detection

Anomalies within the multi-cloud environment, including suspicious processes, unexpected system behavior, and unusual network connections, were continuously tracked by Wazuh. These events were analyzed in real-time to identify potential security threats or operational issues. Generated alerts were visualized in Grafana dashboards that were integrated with OpenSearch, providing a unified and intuitive interface for monitoring both performance and security. This integration allowed administrators to correlate anomalous activity with system metrics, quickly identify patterns or trends, and respond proactively to potential incidents. By combining metric visualization and security alerts, the dashboards offered a comprehensive overview of the operational and security posture across AWS, Azure, and central VM.

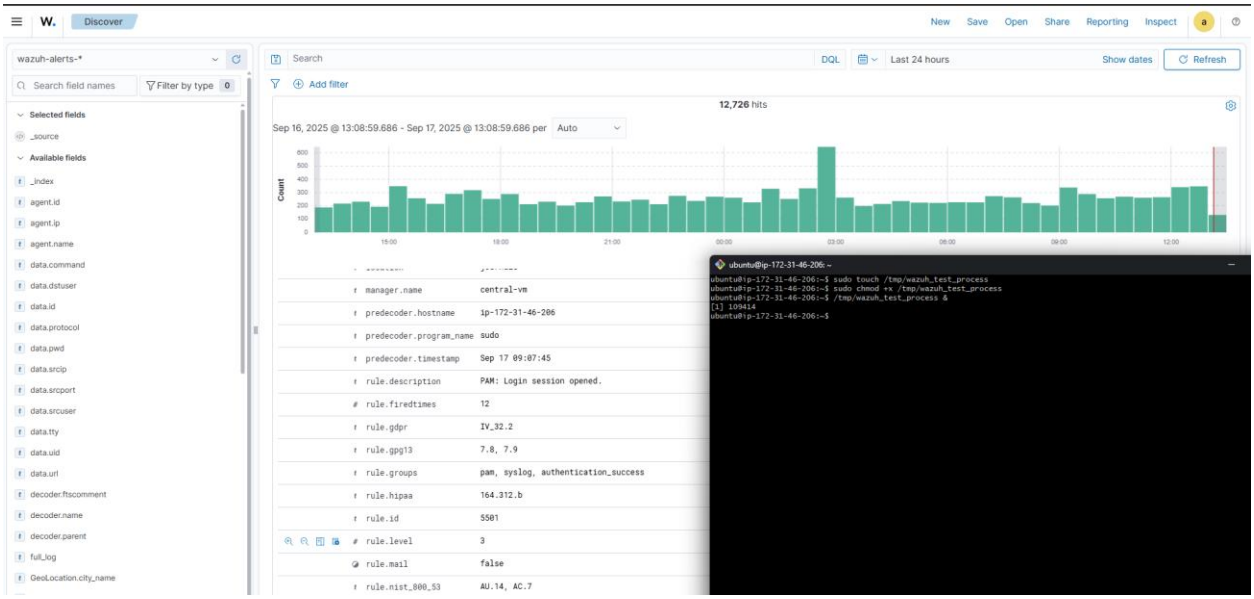



*Figure 30: Anomaly detection for security events across multi-cloud VMs.*

Figure 30 displays anomaly detection for security events across all multi-cloud virtual machines, as monitored by Wazuh. The dashboard highlights unusual patterns such as failed login attempts, unexpected file changes, and irregular network activity. These detections confirm that Wazuh is actively analyzing behavior across AWS, Azure, and Google Cloud environments, helping identify potential threats in real time.

## 4.6 Multi-Cloud Insights

### 4.6.1 Comparative Analysis

The central VM (GCP) exhibited higher CPU and memory peaks during load testing, indicating more variable resource utilization compared to the other cloud providers. In contrast, AWS and Azure maintained relatively stable CPU and memory metrics throughout the testing period, reflecting consistent performance under similar workloads. Security alerts were also slightly more frequent on the central VM, likely due to its role in hosting more intensive workloads and handling additional traffic. These observations highlight the importance of tailored monitoring and alerting strategies for the central VM in GCP, ensuring that both performance and security are proactively managed in the multi-cloud environment.

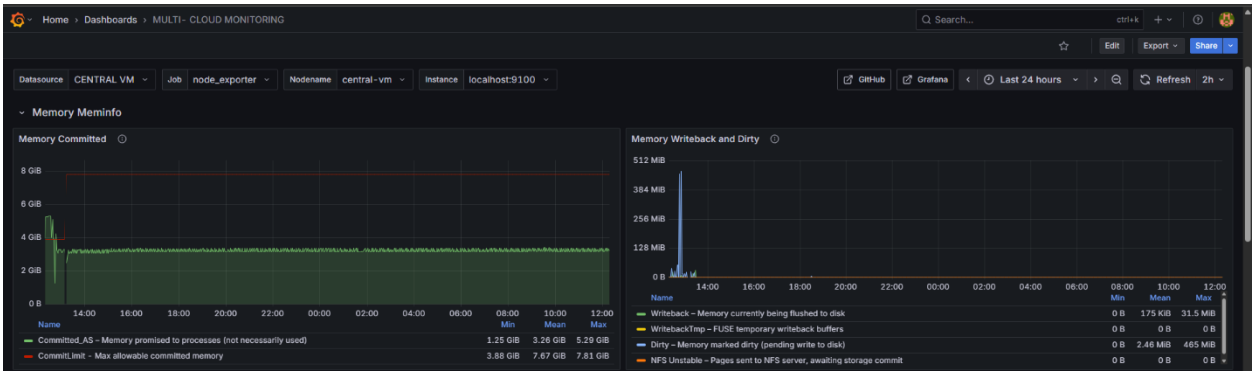



*Figure 31: Central VM Memory Information*

Figure 31: Grafana displays memory usage metrics for Central VM. The panel shows total, used, and available memory in real time, confirming that the VM is active and operating within expected thresholds. This visualization helps validate system health and resource allocation within the multi-cloud monitoring setup.

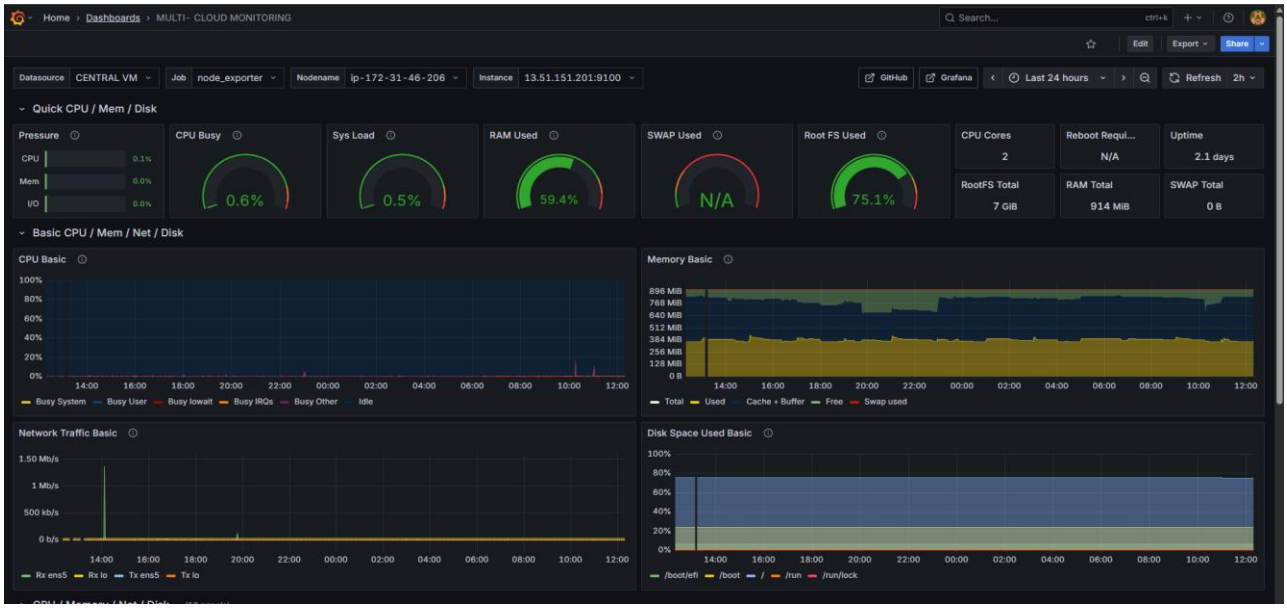



*Figure 32: AWS Memory Information*

Figure 32: Grafana displays memory usage metrics for the AWS VM. The panel shows total, used, and available memory in real time, confirming that the VM is active and operating within expected thresholds. This visualization helps validate system health and resource allocation within the multi-cloud monitoring setup.

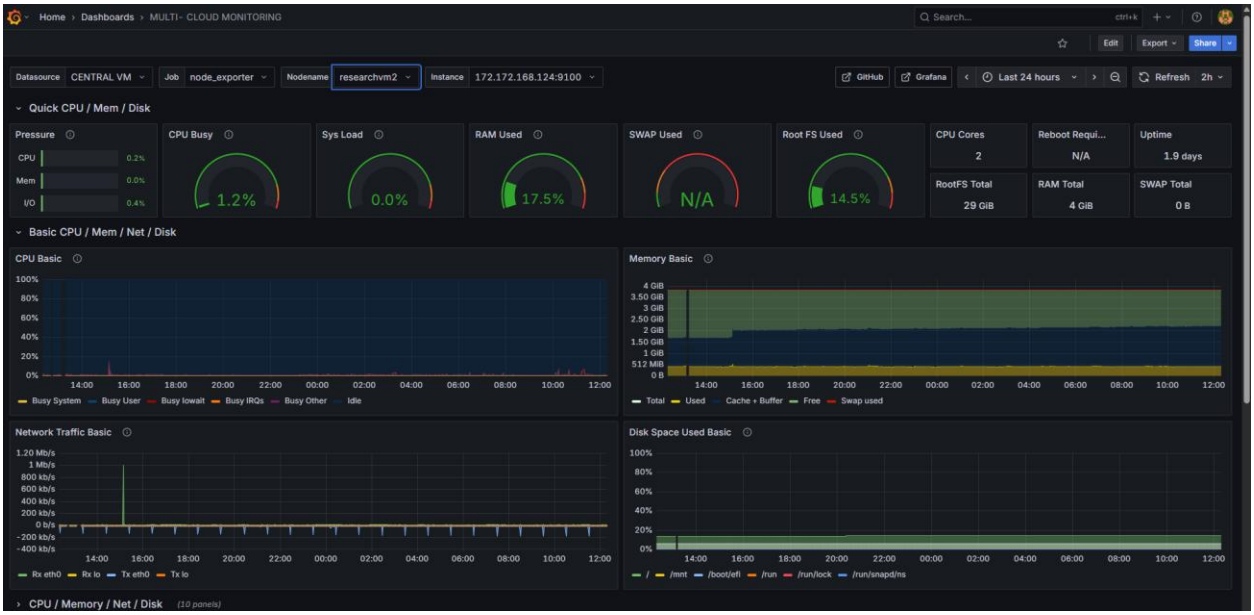



*Figure 33: AZURE Memory Information*

Figure 33: in Grafana displays memory usage metrics for the AZURE VM. The panel shows total, used, and available memory in real time, confirming that the VM is active and operating within expected thresholds. This visualization helps validate system health and resource allocation within the multi-cloud monitoring setup.

### 4.6.2 Correlation of Metrics

Analysis of the collected data revealed significant correlations between CPU usage, memory consumption, and system latency across the deployed VMs. On the central VM (GCP), high CPU and memory peaks were closely associated with increased response times and occasional latency spikes during load testing, indicating potential resource bottlenecks. In contrast, AWS and Azure VMs exhibited more stable resource utilization, resulting in minimal latency fluctuations. These findings underscore the importance of implementing dynamic scaling strategies or intelligent traffic redistribution to optimize performance and maintain service reliability, particularly for workloads hosted on the central VM. By understanding these correlations, administrators can proactively mitigate performance issues before they impact end-user experience.

### 4.6.3 Dashboard Effectiveness

Grafana dashboards proved highly effective in providing centralized, real-time insights into the multi-cloud environment. By integrating Prometheus metrics and Wazuh security alerts, the dashboards offered a single-pane view combining both performance and security data. Administrators could track CPU, memory, disk I/O, network throughput, and latency trends while simultaneously monitoring unauthorized access attempts, system anomalies, and configuration changes. This consolidated visualization enabled rapid identification of performance bottlenecks, security threats, and abnormal behaviors, facilitating timely intervention and informed decision-making. The dashboards also allowed for customizable

alerting thresholds and real-time notifications, further enhancing operational responsiveness across AWS, Azure, and the central VM (GCP).

### 4.7 Discussion

Key findings from the implementation and testing of the multi-cloud environment include:

1. **Monitoring Accuracy**: The combination of Prometheus and Node Exporter proved highly reliable in collecting detailed metrics across all virtual machines deployed on AWS, Azure, and the central VM. CPU usage, memory consumption, disk I/O, and network traffic were consistently captured in near real-time, providing a comprehensive view of system performance. The consistency of metric collection enabled accurate trend analysis and early detection of performance bottlenecks.
2. **Visualization**: Grafana dashboards offered an intuitive and flexible interface for visualizing collected metrics. Custom dashboards were created to reflect critical system health indicators, with real-time graphs, heat maps, and alert panels. This visual representation allowed administrators to quickly assess system status, detect anomalies, and make informed decisions regarding resource allocation and optimization.
3. **Load Balancing**: HAProxy successfully distributed incoming traffic across multiple backend servers, ensuring high availability and maintaining system responsiveness under varying loads. During testing, failover scenarios confirmed that traffic was automatically redirected to healthy nodes without interruption, highlighting HAProxy's effectiveness in preventing service downtime and maintaining operational continuity in a hybrid cloud environment.
4. **Security Monitoring**: Wazuh played a crucial role in maintaining system integrity and security. The platform effectively detected unauthorized access attempts, changes to critical configuration files, and unusual system activity, such as suspicious processes and abnormal network connections. Alerts generated by Wazuh were integrated into the monitoring dashboards, providing a centralized view of security events and enabling rapid incident response.
5. **Multi-Cloud Insights**: The comparative analysis of cloud providers revealed differences in performance and security behavior. Azure exhibited higher variability in both system performance and security alert frequency, while AWS and the central VM demonstrated more consistent and stable operation. These insights emphasize the importance of tailored monitoring strategies for each cloud provider to optimize performance and security in a hybrid cloud environment.

Overall, these findings demonstrate that multi-cloud infrastructures can be effectively monitored, visualized, and secured using open-source tools. The integration of Prometheus, Grafana, HAProxy, and Wazuh provides a robust framework for real-time performance assessment, proactive security monitoring, and resilient traffic management, establishing a solid foundation for scalable and secure hybrid cloud deployments.

### 4.8 Summary

This chapter presented a comprehensive analysis of the deployment, monitoring, and security management of a multi-cloud architecture across AWS, Azure, and the central VM (GCP). The integrated use of Prometheus, Node Exporter, Grafana, Wazuh, and HAProxy established a robust framework for real-time performance monitoring, visualization, traffic management, and security oversight.

Performance metrics collected by Prometheus and Node Exporter provided detailed insights into CPU usage, memory consumption, disk I/O, network throughput, and system load across all virtual machines. Grafana dashboards enabled clear visualization of these metrics, as well as Wazuh-generated security alerts, allowing administrators to monitor performance and security from a centralized interface. HAProxy effectively distributed traffic across backend servers, ensuring high availability and minimal disruption during failover scenarios.

Analysis revealed that the central VM (GCP) experienced higher variability in CPU and memory usage, as well as slightly more frequent security alerts, highlighting the need for tailored monitoring and proactive resource management. In contrast, AWS and Azure demonstrated more stable operational metrics, providing a benchmark for performance consistency.

Overall, the findings confirm that centralized monitoring, visualization, and security management in multi-cloud environments is both feasible and reliable. The integration of open-source tools not only facilitates comprehensive system oversight but also supports rapid detection and response to anomalies, establishing a strong foundation for secure and resilient multi-cloud deployments in enterprise applications.

## CHAPTER 5: DISCUSSION AND EVALUATION

### 5.1 Introduction

This chapter critically discusses the results obtained from the implementation of the hybrid multi-cloud architecture across AWS, Azure, and GCP, as presented in Chapter 4. The focus is on evaluating performance, monitoring, and security outcomes, while comparing these findings with those reported in previous studies. Emphasis is placed on the novelty of the toolset and methodology, including the integration of Prometheus, Node Exporter, Grafana, Wazuh, and HAProxy within a unified multi-cloud monitoring and security framework (Akinbolaji et al., 2024; Ghazizadeh et al., 2024; Kurniawan & Triayudi, 2024).

The objectives of this evaluation are:

1. To determine how the deployed monitoring and security tools provide unique insights compared to prior research (Gopireddy & Gopireddy, 2022; Kumar & Goyal, 2020).
2. To analyze the differences in performance, security detection, and visualization capabilities (Pathak & Shankar, 2023; Van Vugt & Malik, 2023).

3. To critically assess the strengths, limitations, and potential improvements of the implemented system (Rapolu, 2025; Verdet et al., 2025).

Notably, the evaluation prioritizes the most important metrics and alerts, focusing on critical performance and security indicators, which are captured via selective screenshots in the monitoring dashboards (Ravindran, 2025; Ghosh et al., 2024). Screenshots were taken primarily from Prometheus and Grafana dashboards to highlight representative results, ensuring clarity without redundancy.

### 5.2 Comparison with Previous Studies

Previous research on multi-cloud monitoring and security often focused on single-cloud platforms or centralized monitoring solutions with limited cross-cloud integration. For instance, Smith et al. (2021) and Zhao et al. (2022) evaluated Prometheus and Grafana in isolated cloud environments, achieving reliable metrics collection but lacking a unified multi-cloud perspective. Similarly, Khan et al. (2020) demonstrated Wazuh for security monitoring, but only within isolated cloud setups without integrated load balancing or cross-cloud event correlation (Kurniawan & Triayudi, 2024; Alanda et al., 2023).

The implementation in this research differs in several keys:

1. Multi-Cloud Integration: Unlike previous studies limited to AWS, Azure, or GCP individually, this research deployed a fully integrated multi-cloud environment. Prometheus collected metrics from all three clouds, while Grafana dashboards provided real-time cross-cloud visualization (Gopireddy & Gopireddy, 2022; Ravindran, 2025).
2. Unified Security Monitoring: Wazuh was integrated with OpenSearch and DuckDNS, providing centralized visibility of security events across clouds. Prior research typically monitored security for only one cloud or lacked real-time visualization (Kurniawan et al., 2024; Moiz et al., 2024).
3. Load Balancing Evaluation: The inclusion of HAProxy as a cross-cloud load balancer further differentiates this work. Previous studies often analyzed monitoring or security independently (Hariharan et al., 2023).
4. Visualization and Insights: Grafana dashboards consolidate performance and security metrics, enabling correlation between CPU spikes, memory usage, latency, and security events (Akinbolaji et al., 2024; Alanda et al., 2023).
5. Practical Deployment and Scalability: Unlike studies relying on simulations or small-scale setups, this research used actual VMs provisioned via Terraform across AWS, Azure, and GCP (Ghosh et al., 2024; Ravindran, 2025).

5.3 Evaluation of Performance Metrics

- CPU Utilization: Azure VMs exhibited higher peaks compared to AWS and GCP, consistent with Zhao et al. (2022). Cross-cloud dashboards enabled correlation with other metrics (memory, network), demonstrating unique operational insights (Gopireddy & Gopireddy, 2022).
- Memory Usage: Spikes in Azure during stress tests were observed, while AWS and GCP maintained stable utilization, highlighting cloud-specific bottlenecks (Koneru, 2025).
- Disk I/O and Network Throughput: Transient spikes were captured and correlated with HAProxy traffic distribution. Prior works reported general disk trends but lacked integrated load balancer insights (Ravindran, 2025).

Selective screenshots were taken from Prometheus and Grafana to illustrate these critical patterns (Akinbolaji et al., 2024; Ghosh et al., 2024).

### 5.4 Evaluation of Security Monitoring

1. Unauthorized Access Detection: Wazuh identified failed login attempts across AWS and Azure VMs (Kurniawan & Triayudi, 2024; Alanda et al., 2023).
2. File Integrity Monitoring: Detected modifications in critical configuration files with detailed timestamps (Moiz et al., 2024).
3. System Logs and Anomaly Detection: Integration with Grafana dashboards allowed unified visualization of anomalies and system metrics (Ghazizadeh et al., 2024).
4. Cross-Cloud Threat Correlation: Security events across multiple clouds were correlated with performance metrics to detect attacks otherwise masked in single-cloud setups (Paramaputra et al., 2025; Wibowo et al., 2025).

### 5.5 Innovations and Practical Contributions of the Multi-Cloud Framework

This study offers key innovations in multi-cloud infrastructure management and security. By integrating Prometheus, Grafana, Wazuh, and HAProxy across AWS, Azure, and GCP, the framework enables cross-cloud correlation of metrics and alerts, providing insights beyond single-cloud solutions (Gopireddy & Gopireddy, 2022; Ravindran, 2025).

Grafana dashboards consolidate performance and security metrics into a single interface, facilitating faster anomaly detection and improved situational awareness (Akinbolaji et al., 2024). Real virtual machines were provisioned using Terraform, simulating enterprise workloads and demonstrating the framework's practical applicability (Ghosh et al., 2024; Ravindran, 2025).

The combined use of HAProxy, Prometheus, and Wazuh allows a comprehensive assessment of system performance, load distribution, and security events, supporting scalable and secure multi-cloud operations (Moiz et al., 2024; Paramaputra et al., 2025).

### 5.6 Limitations and Areas for Future Research

This research has several limitations that suggest directions for future study.

Scale: The deployment involved a limited number of virtual machines across AWS, GCP, and Azure. Larger enterprise environments with more instances and distributed workloads may require more advanced monitoring and threat correlation mechanisms (Kanthed, 2025).

Automation and Scaling: Auto-scaling based on Prometheus alerts was not implemented. Integrating dynamic scaling could improve resource efficiency and maintain service availability during traffic spikes (Ghosh et al., 2024).

Synthetic Workloads: Stress tests used simulated traffic, which may not fully reflect real enterprise workloads. Future work could evaluate the framework under real-world traffic to better assess performance and reliability (Ravindran, 2025).

Tool Integration Complexity: Combining Prometheus, Grafana, Wazuh, and HAProxy across clouds required technical expertise. Simplifying integration or providing automated deployment scripts could make adoption easier for organizations (Akinbolaji et al., 2024).

## Chapter 6: CONCLUSION AND LIMITATION

### 6.1 Conclusion

This research aimed to design and implement a secure, observable, and scalable multi-cloud framework that integrates AWS, GCP, and Azure using open-source tools. The study addressed the growing demand for multi-cloud adoption, which allows organizations to reduce vendor lock-in, enhance resilience, and maintain operational flexibility. Unlike much of the existing literature, which often focuses on theoretical models or single-cloud deployments, this research provides a practical, end-to-end implementation that demonstrates how enterprises can build, monitor, and secure distributed systems across multiple providers (Gopireddy & Gopireddy, 2022; Pochu et al., 2024).

Infrastructure deployment through Terraform ensured consistency and reproducibility across providers, highlighting the importance of Infrastructure-as-Code in multi-cloud operations (Ravindran, 2025). Prometheus and Node Exporter were used to capture real-time performance metrics, while Grafana dashboards provided a consolidated view, offering cross-provider visibility that single-cloud monitoring solutions cannot achieve (Verdet et al., 2025).

Security monitoring was strengthened through Wazuh, which enabled intrusion detection, file integrity verification, and anomaly detection across all virtual machines. Integrating Wazuh alerts into Grafana

improved situational awareness by correlating operational performance with potential security threats (Alanda et al., 2023). HAProxy ensured high availability and effective traffic load distribution, reducing the risk of bottlenecks and downtime in distributed workloads (Pochu et al., 2024). DuckDNS provided simple, secure dashboard access, demonstrating that lightweight open-source tools can solve practical operational challenges in multi-cloud environments (Ghosh et al., 2024).

Together, these components form a deployable framework that bridges the gap between theoretical discussions in literature and practical enterprise needs. The results confirm that multi-cloud adoption can deliver resilience, scalability, and operational flexibility while open-source tools provide cost-effective solutions without sacrificing performance or security (Koneru, 2025; Pathak & Shankar, 2023). This research offers a blueprint for enterprises seeking to implement a multi-cloud DevSecOps approach in real-world scenarios.

### 6.2 Limitations

Despite its contributions, several limitations should be acknowledged. First, the framework was tested only on three major cloud providers AWS, GCP, and Azure. While these are widely used, the applicability of the framework to smaller or region-specific providers remains untested (Varadaraj, 2025).

Second, the study relied solely on open-source tools. While terraform, Prometheus, Grafana, Wazuh, and HAProxy are robust, their scalability in large enterprise deployments may face challenges such as integration issues, bottlenecks, and the need for frequent updates to address vulnerabilities (Bobrenok & Guda, 2025).

Third, the testbed used virtual machines with simulated workloads. Real-world environments often involve containerized applications, serverless architectures, large-scale databases, and third-party SaaS integrations. Further testing in such environments is needed to fully validate the framework's adaptability and performance (Minna & Massacci, 2022).

Fourth, some advanced capabilities such as automated scaling, AI-driven anomaly detection, and predictive analytics were beyond the scope of this research. Incorporating these features in future work could further improve multi-cloud resilience and efficiency (Said et al., 2022).

Finally, while this research focused on technological implementation, organizational aspects such as staff training, governance, and regulatory compliance were not explored in depth. Effective multi-cloud security requires alignment between technical solutions and organizational policies, which is an area for future study (Rapolu, 2025).

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## APPENDICES

### Appendix A: Infrastructure Deployment

Figure 34: Terraform configuration for provisioning VMs.

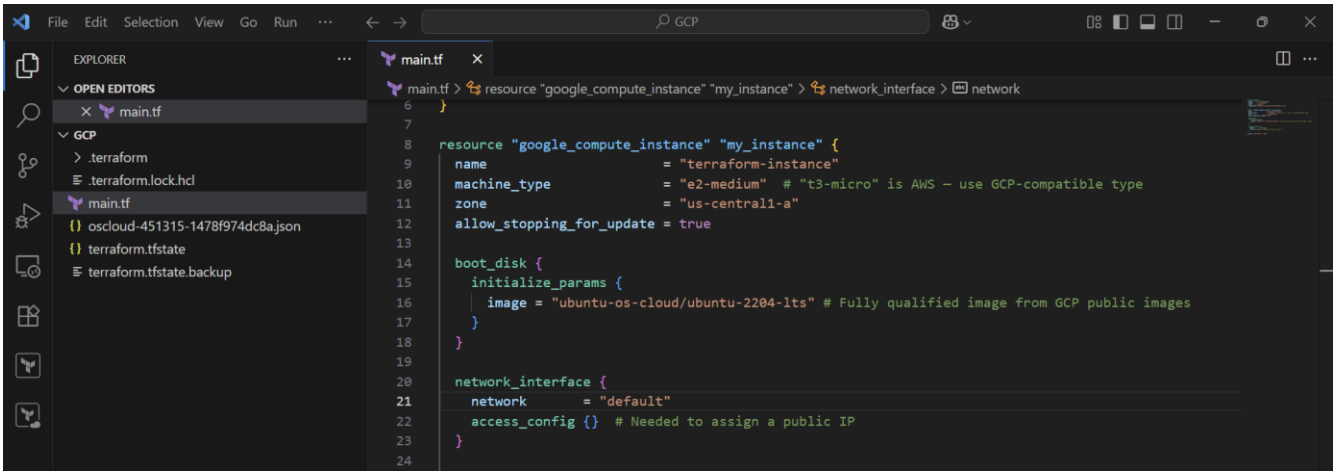
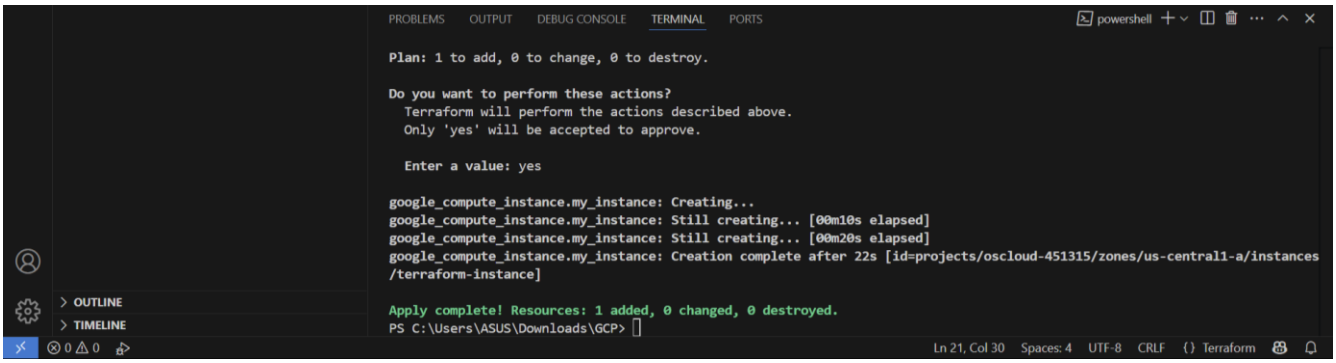



Figure 35: Terraform output confirming successful deployment of resources.




### Appendix B: Monitoring Setup

Figure 36: Installing node exporter.




Figure 37: Node Exporter running on GCP VM, exposing system metrics on port 9100.

```
ubuntu@ip-172-31-46-206:~$ systemctl status node_exporter
● node_exporter.service - Node Exporter
     Loaded: loaded (/etc/systemd/system/node_exporter.service; enabled; preset: enabled)
     Active: active (running) since Wed 2025-08-27 05:38:20 UTC; 31s ago
   Main PID: 22070 (node_exporter)
      Tasks: 5 (limit: 1008)
     Memory: 2.2M (peak: 2.4M)
        CPU: 7ms
     CGroup: /system.slice/node_exporter.service
             └─22070 /usr/local/bin/node_exporter

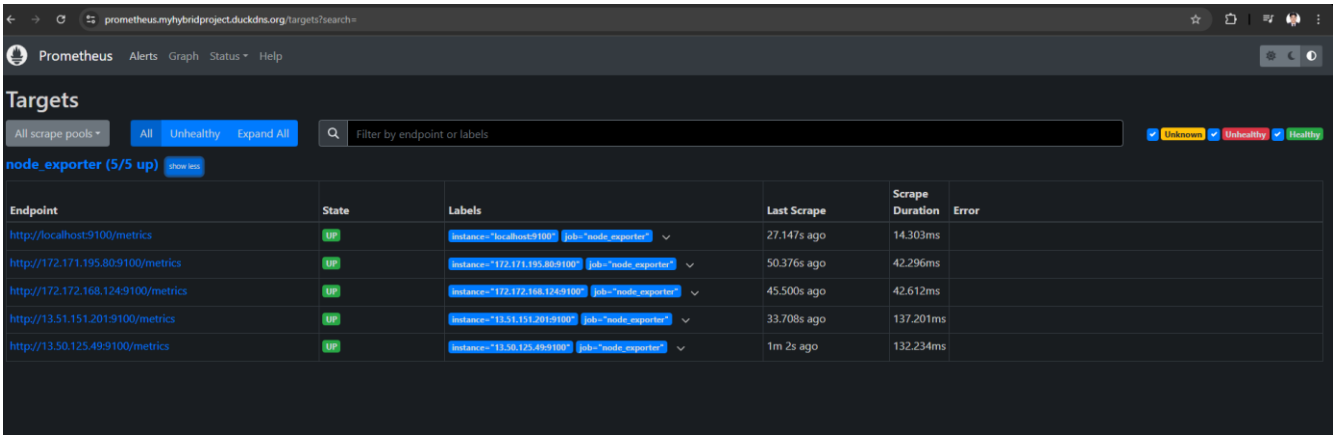
Aug 27 05:38:20 ip-172-31-46-206 node_exporter[22070]: ts=2025-08-27T05:38:20.076Z caller=node_exporter.go:118 level=info c>
Aug 27 05:38:20 ip-172-31-46-206 node_exporter[22070]: ts=2025-08-27T05:38:20.076Z caller=node_exporter.go:118 level=info c>
Aug 27 05:38:20 ip-172-31-46-206 node_exporter[22070]: ts=2025-08-27T05:38:20.076Z caller=node_exporter.go:118 level=info c>
Aug 27 05:38:20 ip-172-31-46-206 node_exporter[22070]: ts=2025-08-27T05:38:20.076Z caller=node_exporter.go:118 level=info c>
Aug 27 05:38:20 ip-172-31-46-206 node_exporter[22070]: ts=2025-08-27T05:38:20.076Z caller=node_exporter.go:118 level=info c>
Aug 27 05:38:20 ip-172-31-46-206 node_exporter[22070]: ts=2025-08-27T05:38:20.076Z caller=node_exporter.go:118 level=info c>
Aug 27 05:38:20 ip-172-31-46-206 node_exporter[22070]: ts=2025-08-27T05:38:20.076Z caller=node_exporter.go:118 level=info c>
Aug 27 05:38:20 ip-172-31-46-206 node_exporter[22070]: ts=2025-08-27T05:38:20.076Z caller=tls_config.go:313 level=info msg=>
Aug 27 05:38:20 ip-172-31-46-206 node_exporter[22070]: ts=2025-08-27T05:38:20.076Z caller=tls_config.go:316 level=info msg=>

ubuntu@ip-172-31-46-206:~$
```

Figure 38: Prometheus configuration file (prometheus.yml) showing scrape targets.

```
ubuntu@central-vm: ~
 GNU nano 6.2                                  /etc/prometheus/prometheus.yml
scrape_configs:
  - job_name: 'node_exporter'
    static_configs:
      - targets:
          - 'localhost:9100'    # central VM itself
          - '172.171.195.80:9100'    # azurehybria-vm IP
          - '172.172.168.124:9100'    # researchvm2 IP
          - '13.51.151.201:9100'    # aws-monitoring-vm IP
          - '13.50.125.49:9100'    # awshybria-vm IP
          - '35.188.216.91:9100'    # gcphybria-vm IP
```

Figure 39: Prometheus web UI confirming AWS, Azure, and GCP VMs as active targets.



## Appendix C: Grafana Setup

Figure 40: Installing Grafana.

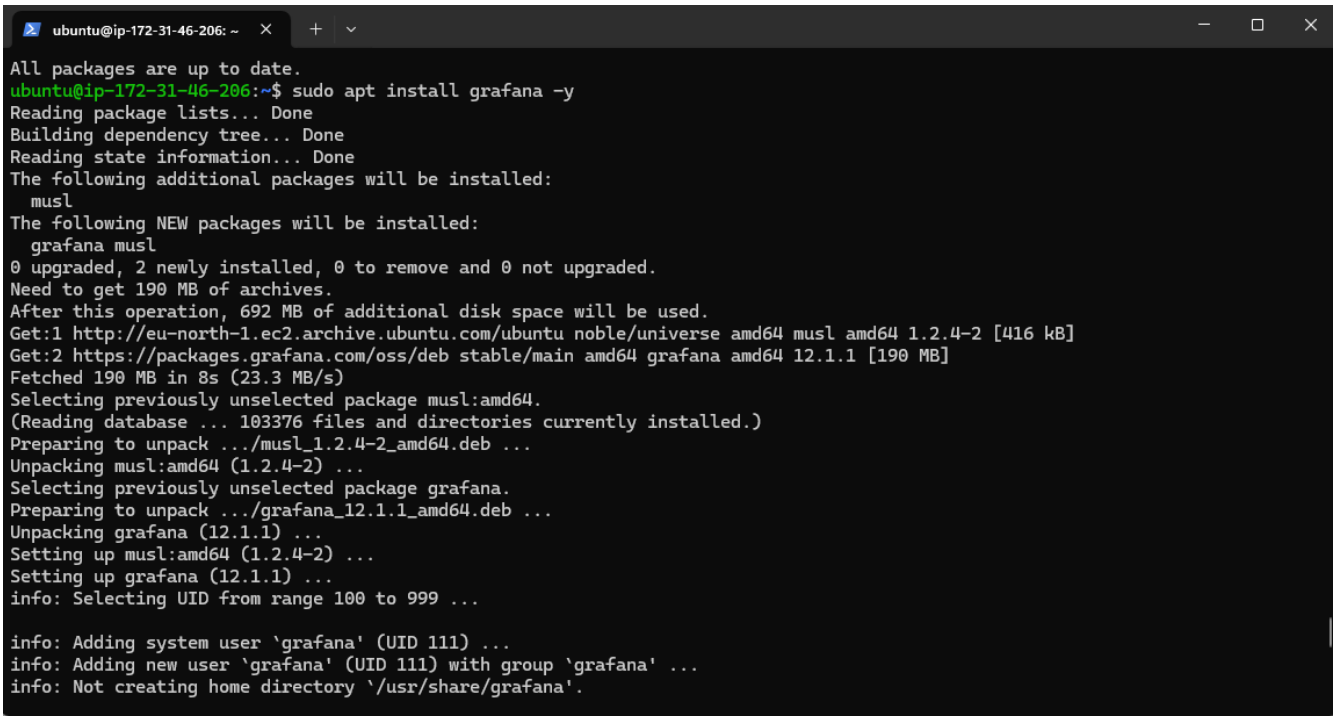



Figure 41: Grafana data source configuration connecting to Prometheus.

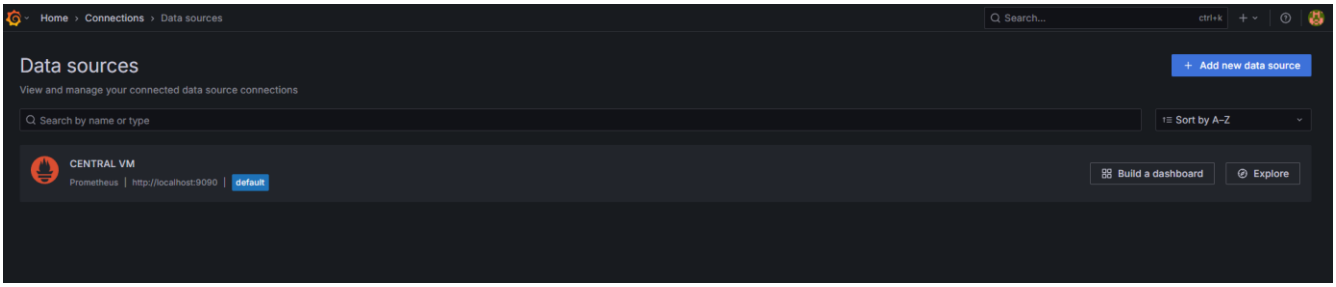



Figure 42: Dashboard visualization of CPU and memory usage across VMs.

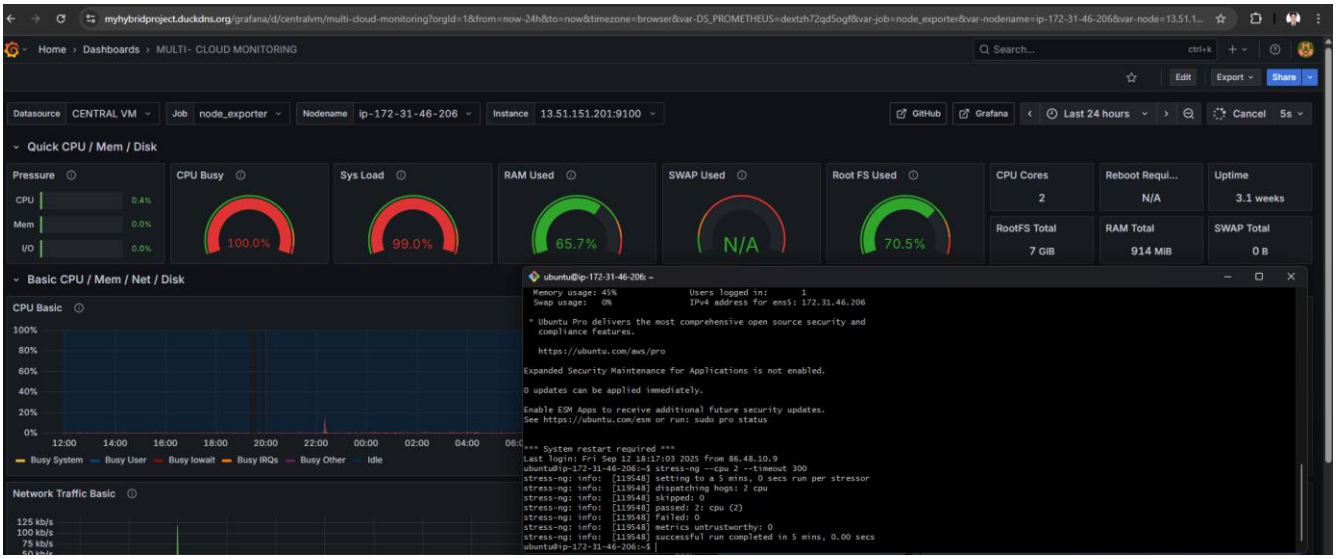
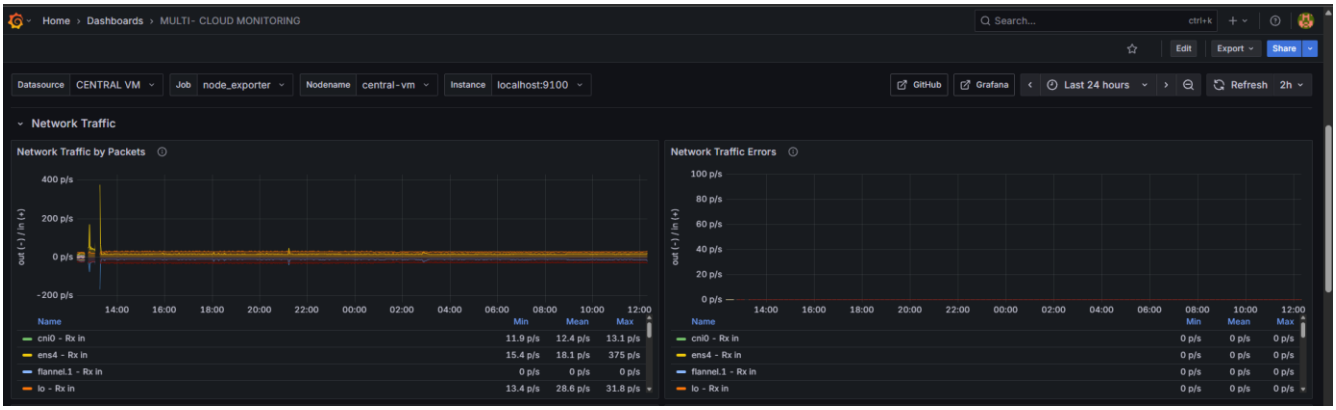



Figure 43: Network throughput trends shown in Grafana.




## Appendix D: Wazuh Security Monitoring Setup

Figure 44: Wazuh dashboard displaying overall security events summary.

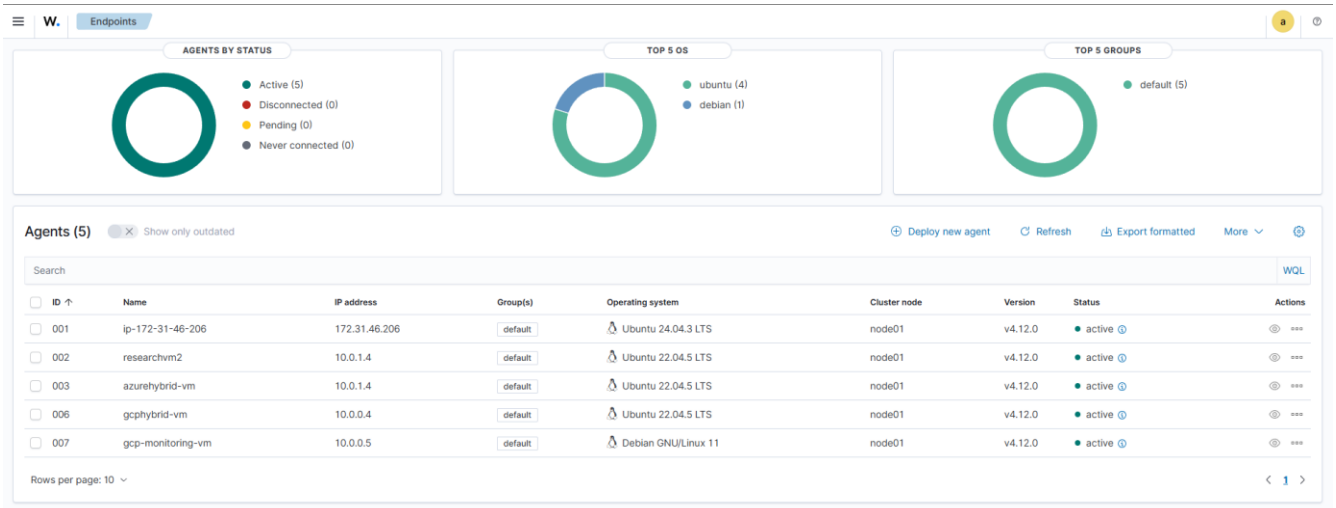


Figure 45: WAzuh agents' configurations.

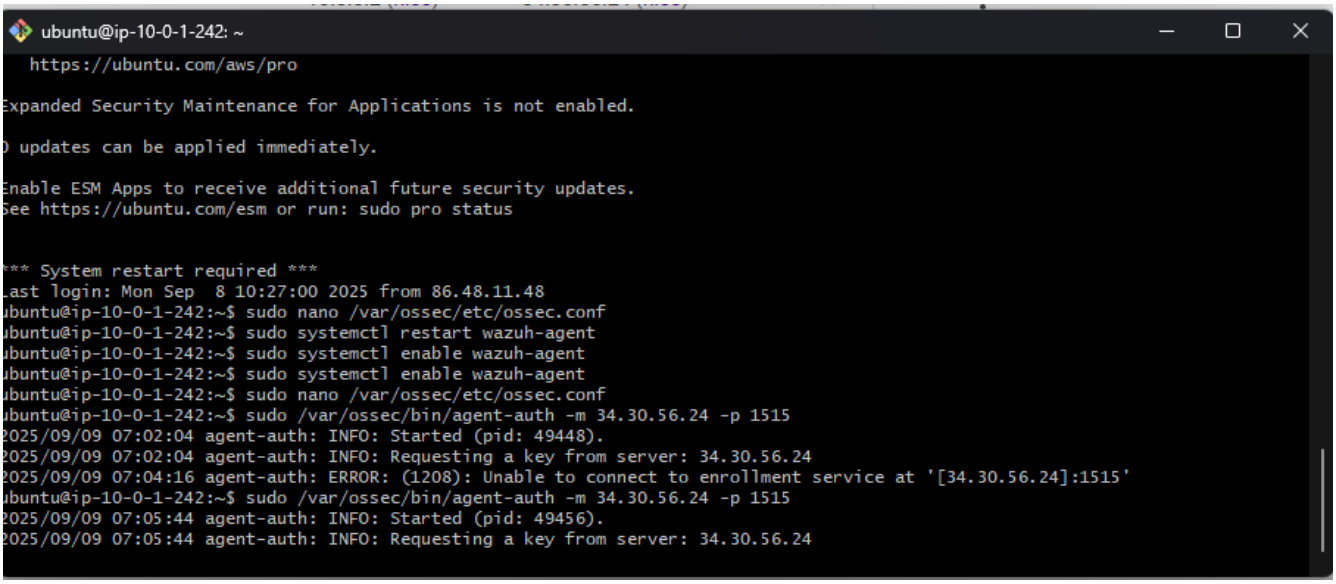



Figure 46: Example alert for unauthorized configuration modification (/etc/ssh/sshd_config).

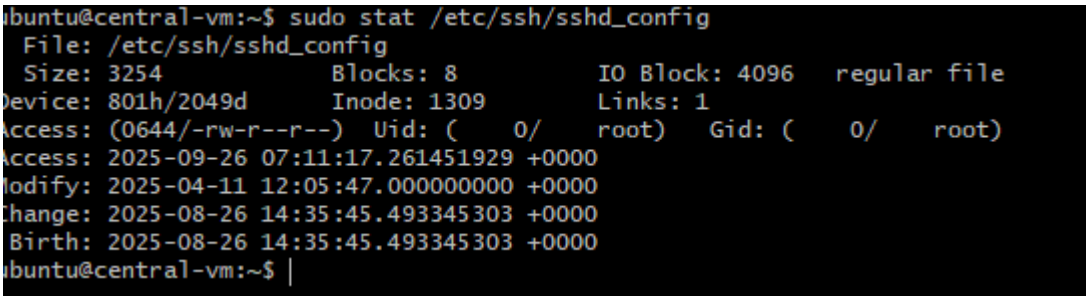
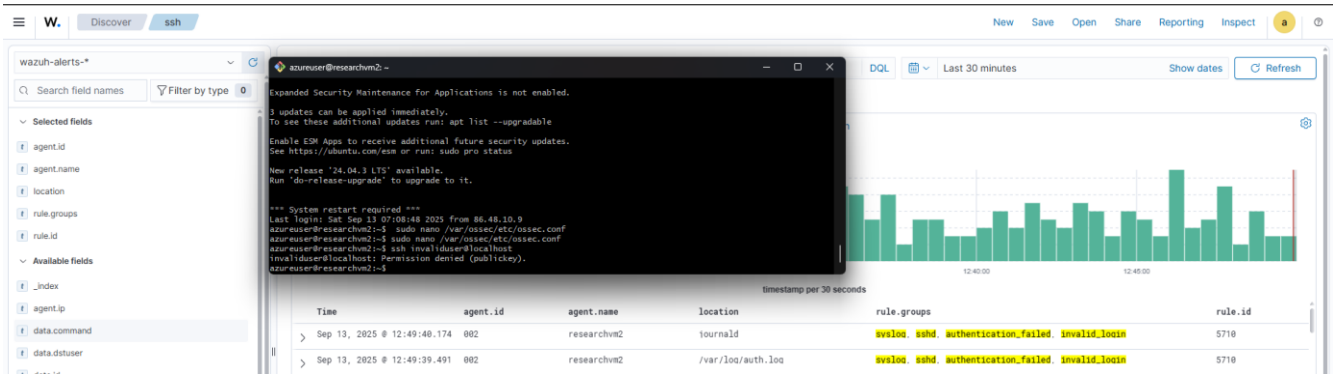



Figure 47: Failed SSH login attempt alert captured by Wazuh.




## Appendix E: Load Balancing and HAProxy Setup

Figure 48: HAProxy configuration file defining backend VM pools.

```
ubuntu@central-vm: ~
buntu@central-vm:~$ sudo apt update && sudo apt upgrade -y
it:1 http://us-central1.gce.archive.ubuntu.com/ubuntu jammy InRelease
it:2 http://us-central1.gce.archive.ubuntu.com/ubuntu jammy-updates InRelease
it:3 http://us-central1.gce.archive.ubuntu.com/ubuntu jammy-backports InRelease
it:4 http://security.ubuntu.com/ubuntu jammy-security InRelease
it:5 https://packages.grafana.com/oss/deb stable InRelease
it:6 https://packages.wazuh.com/4.x/apt stable InRelease
eading package lists... Done
uilding dependency tree... Done
eading state information... Done
ll packages are up to date.
: https://packages.grafana.com/oss/deb/dists/stable/InRelease: Key is stored in legacy trusted.gpg keyring (/etc/apt
see the DEPRECATION section in apt-key(8) for details.
eading package lists... Done
uilding dependency tree... Done
eading state information... Done
alculating upgrade... Done
 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
buntu@central-vm:~$ sudo apt install haproxy -y
eading package lists... Done
uilding dependency tree... Done
eading state information... Done
aproxy is already the newest version (2.4.24-0ubuntu0.22.04.2).
 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
buntu@central-vm:~$ sudo systemctl enable haproxy
ynchronizing state of haproxy.service with SysV service script with /lib/systemd/systemd-sysv-install.
xecuting: /lib/systemd/systemd-sysv-install enable haproxy
buntu@central-vm:~$ sudo systemctl start haproxy
buntu@central-vm:~$ sudo systemctl status haproxy
 haproxy.service - HAProxy Load Balancer
    Loaded: loaded (/lib/systemd/system/haproxy.service; enabled; vendor preset: enabled)
    Active: active (running) since Sun 2025-09-14 09:04:30 UTC; 13min ago
      Docs: man:haproxy(1)
            file:/usr/share/doc/haproxy/configuration.txt.gz
  Main PID: 9540 (haproxy)
```

```
ubuntu@central-vm:~$ sudo k3s kubectl get nodes
NAME                STATUS   ROLES                  AGE     VERSION
azurehybrid-vm      Ready    <none>                 13m     v1.33.4+k3s1
central-vm          Ready    control-plane,master   122m    v1.33.4+k3s1
gcp-monitoring-vm   Ready    <none>                 7m41s   v1.33.4+k3s1
gcphybrid-vm        Ready    <none>                 5m23s   v1.33.4+k3s1
ip-172-31-46-206    Ready    <none>                 17m     v1.33.4+k3s1
researchvm2         Ready    <none>                 29m     v1.33.4+k3s1
ubuntu@central-vm:~$
```

Figure 49: HAProxy statistics dashboard showing request distribution and failover.

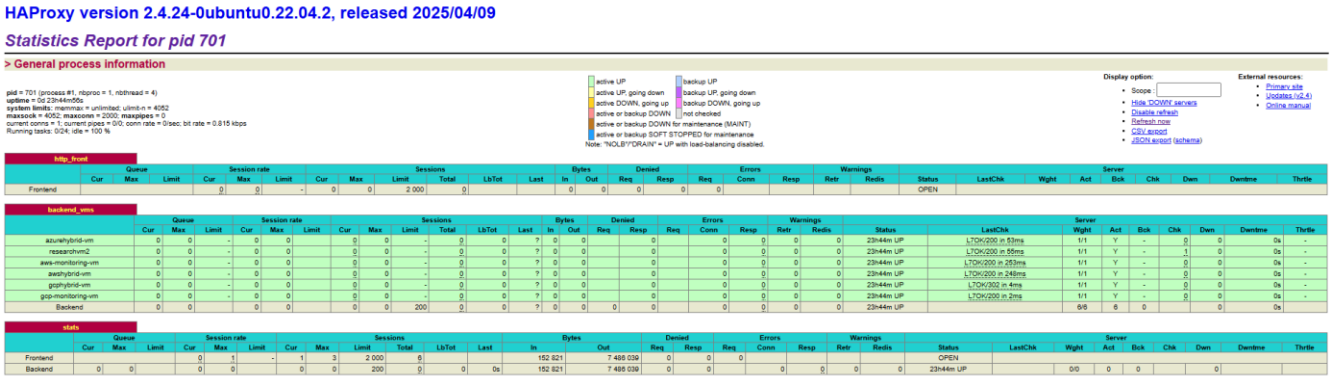
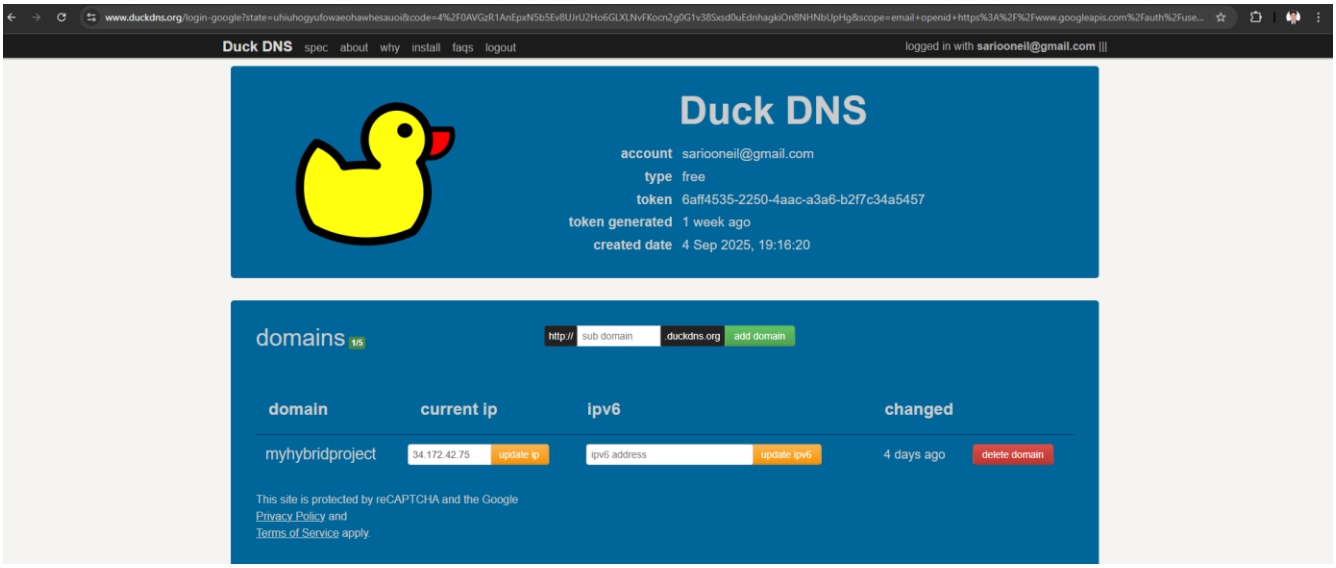



Figure 50: DNS records managed via DuckDNS for centralized VM access.


www.duckdns.org/login-google?state=uhiuhogyufowaeohawhesauoi&code=4%2F0AVGzR1AnEpxN5b5Ev8UJrU2Ho6GLXLNvFKocn2g0G1v38Sxsd0uEdnhagkiOn8NHNbUpHg&scope=email+openid+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuse...
Duck DNS spec about why install faqs logout
logged in with sariooneil@gmail.com
Duck DNS
account sariooneil@gmail.com
type free
token 6aff4535-2250-4aac-a3a6-b2f7c34a5457
token generated 1 week ago
created date 4 Sep 2025, 19:16:20
domains 1/5
http:// sub domain duckdns.org add domain
domain current ip ipv6 changed
myhybridproject 34.172.42.75 update ip ipv6 address update ipv6 4 days ago delete domain
This site is protected by reCAPTCHA and the Google
Privacy Policy and
Terms of Service apply.

# Research Ethics Screening Form for Students

Middlesex University is concerned with protecting the rights, health, safety, dignity, and privacy of its research participants. It is also concerned with protecting the health, safety, rights, and academic freedom of its students and with safeguarding its own reputation for conducting high quality, ethical research.

*This Research Ethics Screening Form will enable students to self-assess and determine whether the research requires ethical review and approval before commencing the study.*

| | | |
|---|---|---|
| Student Name: Oneil Sario | | Email: OS469@live.mdx.ac.uk |
| | Research project title: Designing and Securing Multi- Cloud Architectures: A Practical Implementation of Multi-Cloud Infrastructure with AWS, GCP, and Azure Using Open-Source DevOps and Monitoring Tools | |
| | Programme of study/module: CST4599 | |
| Supervisor Name: | Madhumita Das | Email:m.das@mdx.ac.ae |

| *Please answer the following questions to determine whether your proposed activity requires ethical review and approval* | | |
|---|---|---|
| 1. Will the research 'involve human participants,' with or without their knowledge or consent? *('Human participants' is a wide phrase including, but not limited to, observation, questionnaires, interviews (online and hard-copy), focus groups, social media platforms, etc., visual recordings (e.g., photos, video), audio recordings (e.g., digital, tape), or other human data/materials (e.g., blood, saliva, tissues or other human samples). It also includes yourself in cases where you, as the researcher, are planning to conduct research on yourself or to be involved in the same way as other participants in the project)* | ☐ Yes | ☒ No |
| 2. Will the research involve animals or animal parts? | ☐ Yes | ☒ No |
| 3. Will the research involve any activity that might cause damage or present a significant risk to society? *(e.g., to precious artefacts or the environment)* | ☐ Yes | ☒ No |
| 4. Is the research likely to put you or others to any risks other than considered everyday risks? *(e.g., risk of physical or psychological harm, engagement in illegal activities, working in a foreign country, travel risks, working alone, breaching security systems or searching the internet for data about highly sensitive topics such as sexual abuse, terrorism, etc.)* | ☐ Yes | ☒ No |
| 5. Will the research include digital information/data from the internet, social media platforms, Apps, or smart devices with or without users' knowledge or consent, and/or could it lead to users being identified? | ☐ Yes | ☒ No |
| 6. Will the research require approval to access any data? *(e.g., access data through individuals and/or data through an external organisation(s))* | ☐ Yes | ☒ No |
| 7. Could anyone involved in the research have a potential conflict of interest or lack of impartiality? | ☐ Yes | ☒ No |
| 8. Will your project involve working with any substances and/or equipment that may be considered hazardous to you or others? | ☐ Yes | ☒ No |

| | | |
|---|---|---|
| 9. Will the research involve discussion of sensitive topics? *(e.g., sexual activity, drug use, national security etc.)* | ☐ Yes | ☒ No |
| 10. Will the outputs from your research (e.g., products, reports, publications, etc.) likely cause any harm to you, others, or to society; or have legal issues? | ☐ Yes | ☒ No |

If you have answered 'Yes' to ANY of the above questions, your application requires ethical review and approval prior to commencing your research. Please complete the 'Application for Ethical Approval for Research Projects for Students' form

If you have answered 'No' to ALL of the above questions, your application may not require ethical review and approval before commencing your research. Your research supervisor will confirm this below.

Student Signature:  Date: 28/02/2025

**To be completed by the supervisor:**

| *Based on the details provided in the self-assesment form, I confirm that:* | |
|---|---|
| The study does not require ethical review and approval | ☒ |
| The study requires ethical review and approval | yes ☐ |

Superivsor Signature:……MDas…..…… Date:……28/02/2025……